Why Has the Value of Cash Increased Over Time?

Thomas W. Bates, Ching-Hung Chang, and Jianxin Daniel Chi*

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Abstract:

The value of corporate cash holdings has increased significantly in recent decades. On average, one dollar of cash is valued at $0.61 in the 1980s, $1.04 in the 1990s, and $1.12 in the 2000s. This increase is predominantly driven by the investment opportunity set and cash-flow volatility, as well as secular trends in product market competition, credit market risk, and within-firm diversification. We document a secular decrease in the speed of adjustment in cash holdings, particularly for financially constrained firms with cash deficits, suggesting that capital market frictions can account for the trend in the value of cash holdings.

JEL codes: G30, G32, G34, G35

Keywords: Cash Holdings; Value of Cash; Financing Frictions; Speed of Adjustment; Agency Conflicts

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* Bates, thomas.bates@asu.edu, W. P. Carey School of Business, Arizona State University; Chang, chinghungc@ntu.edu.tw, College of Management, National Taiwan University; and Chi, daniel.chi@unlv.edu, Lee Business School, University of Nevada, Las Vegas. We thank Jarrad Harford (the editor) and an anonymous referee, whose suggestions have significantly improved the paper. For helpful comments and suggestions, we also thank Aziz Alimov, Sreedhar Bharath, Claudia Custodio, Mark Huson, Shane Johnson, Tae-Nyun Kim, Michael Lemmon, Robert Parrino, Xunhua Su, Mike Sullivan, Neng Wang, and seminar participants at the 2012 China International Conference in Finance, the 2012 International Conference on Financial Risk and Corporate Finance Management, and the 2012 Financial Management Association Meeting. Chang gratefully acknowledge financial support from Minister of Science and Technology of Taiwan.

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I. Introduction

The increase in cash holdings of U.S. companies has been a focus of the financial press and policy makers in recent years.[1] For example, President Obama has called on U.S. business leaders to spend their corporate cash holdings to spur economic growth, and members of Congress have debated a tax holiday on the repatriation of the foreign cash holdings of multinational firms.[2] Shareholder concern with the increase in cash holdings stems from two related factors: i) what influences the increasing propensity of public corporations to hold cash, and ii) what is the value of incremental cash holdings for a corporation's investors? The literature has identified that the increase in levels of cash can be largely explained by fluctuations in firm characteristics over time, as well as changes in composition of public firms. For example, Bates, Kahle, and Stulz (2009) conclude that the precautionary motive can explain much of the rise in cash holdings over the last three decades. In this paper we address how investors have valued these increasing cash holdings.

In a frictionless market, the marginal value of a dollar in cash for shareholders should be exactly one dollar, where the cost of obtaining and maintaining that reserve equals its expected benefit. The extant literature documents a variety of frictions that cause significant differences in the marginal value of cash in the cross-section of firms. Faulkender and Wang (2006) note that the variation in the marginal value of cash can be attributed to leverage and the relative tax effects of payout decisions. Denis and Sibilkov (2010) find that the value of cash is higher for financially constrained firms. The literature also documents that the value of cash varies with agency conflicts. For example, Dittmar and Mahrt-Smith (2007) show that the value of cash is increasing in an index of shareholder rights, and is also higher for firms with significant outside blockholders. Chen, Harford, and Lin (2015) conclude that a loss of analyst coverage reduces

[1]For example, see "Companies Shun Investment, Hoard Cash – Reluctance to Spend by Consumers and Businesses Fearful of a Domestic Slowdown Hamstrings Pace of Economic Recovery," by Ben Casselman and Justin Lahard, *The Wall Street Journal*, Sept. 17, 2011, p. A2; "The Growing Corporate Cash Hoard," by Bruce Bartlett, *The New York Times*, Feb. 12, 2013.
[2] "Obama Will Tell Chamber Businesses to Put Cash to Use for Jobs," by Mark Drajem and Mike Dorning, *BusinessWeek*, Feb. 7, 2011.

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the value of cash holdings. Liu and Mauer (2011) identify a negative relation between CEO risk-taking incentives (vega) and the value of cash, and attribute this to stockholder-bondholder conflicts.

To date, the literature has offered only limited evidence on the secular variation in the marginal value of cash and its determinants.[3] While the levels of cash holdings of U.S. industrial firms have risen over time, Bates et al. (2009) document that firms consistently hold less cash than predicted by models based on the precautionary demand for cash, despite well documented improvements in liquidity management, decreases in the cost of hedging, and greater access to bank lines of credit.

We find that the marginal value of cash holdings by U.S. nonfinancial firms has increased substantially over the past three decades. Based on the methodology of Faulkender and Wang (2006), the average additional dollar of cash is valued at \$0.61 during the decade of the 1980s, \$1.04 in the 1990s, and \$1.12 in the 2000s. Over the 30-year period, the correlation between time and the value of cash is 0.59. In a regression of annual estimates of the value of cash on a constant and time, the adjusted R^2 is 32%, and time has a positive and significant coefficient that implies an average annual increase of 2.5 cents in the marginal value of a dollar of cash. When we use an alternative cash valuation model of Pinkowitz and Williamson (2004), we find a similar secular trend. We perform a battery of robustness and endogeneity checks and show that our estimates are unlikely to be driven by extreme stock returns, unobservable expectations of future profitability, or trends in profitability.

The increase in the value of cash obtains across a variety of firm characteristics. For example, for firms in the smallest size quintile, the value of an additional dollar of cash for the 1980s, the 1990s, and the 2000s is \$0.69, \$1.07, and \$1.11, respectively. For the largest size

[3] Bates et al. (2009) estimate the value of cash using the model of Pinkowitz and Williamson (2004). They are the first to document that the value of cash increased from the 1980s to the 1990s. They conclude that agency conflicts are unlikely to explain the systematic increase in cash holdings. Chung, Jung, and Park (2012) also document a secular increase in the value of cash, but model the economic determinants of this trend as time-invariant. Among other substantive differences, this study shows that there is substantial heterogeneity in the economic determinants of the value of cash over time.

quintile, it is \$0.39, \$0.66, and \$0.72. We find a similar secular trend in the value of cash for firms with different investment opportunities, varying financial constraints, and high and low cash-flow volatility.

The persistent uptrend in the value of cash is quite striking in light of the finding by Bates et al. (2009) that the cash ratio for the average U.S. industrial firm more than doubled during the sample period. Much of the popular business press has characterized the rise in cash holdings as excessive, motivating calls for greater corporate payouts to shareholders. The academic literature also suggests that large cash balances can be a symptom of heightened agency conflicts.[4] Our findings suggest, however, that investors have viewed the rising cash holdings positively.

After documenting the secular increase in the value of cash, we turn our focus to identifying the factors that explain this phenomenon. Bates et al. (2009) find that part of the increase in the level of cash can be explained by changes in the composition of listed firms. This effect may also contribute to the time trend in the marginal value of cash because newly listed firms tend to have more valuable investment opportunities and greater cash-flow volatility. Consistent with this, we find that firms that IPO later in our sample period can explain part of the increase in the value of cash: 12% of the increase in the 1990s, and 2% of the increase in the 2000s.

Looking beyond the effect of new listings, the literature provides several theories to explain how changing firm characteristics can affect the demand for, and the value of, cash. We first consider whether firm features associated with the transaction and precautionary demand for liquidity influence the marginal value of cash over time. For ease of exposition, we follow the convention in the literature and examine time-series effects by decade. For the 1990s, the increase in the value of cash is positively correlated with firm investment opportunities and cash-

[4] For example, Dittmar, Mahrt-Smith, and Servaes (2003), among others, show that higher agency conflicts are associated with higher level of cash holdings. Harford (1999) suggests that investors should be concerned with cash-rich firms because these firms tend to make value-destroying acquisitions.

flow volatility. For the 2000s, neither investment opportunities nor cash-flow volatility can explain the increase in the value of cash. Financial constraints exhibit a positive effect on the value of cash, and this effect increases in the 1990s and 2000s.[5]

We extend our models to incorporate the influence of credit market conditions, which varied substantially in the 2000s. Duchin, Ozbas, and Sensoy (2010) find that the credit market supply shock associated with the financial crisis in the late 2000s had the greatest effect on investment for firms with low cash reserves and firms reliant on external capital. We use the Aaa-minus-Baa corporate bond yield to explore whether the change in the value of cash over time is affected by changes in credit spread. We find that for both the 1990s and the 2000s, cash holdings become more valuable when credit spread widens, and this effect is more pronounced in the 2000s. The effect of credit spread is one of the most important determinants of the increase in cash value in the 2000s.[6]

We consider three additional economic forces that have varied substantially during our sample period and are likely to explain the increase in the value of cash. First, as noted by Irvine and Pontiff (2009), product market competition has intensified, and Frésard (2010) finds that cash holdings are strategically valuable for firms operating in competitive product markets. Following the literature, we use a Herfindahl index of market concentration, as well as industry turnover, to estimate the intensity of competition and find that product market competition is a significant determinant of the increase in the value of cash during the 1990s. Second, Duchin (2010) documents that decreasing diversification in multi-segment firms has contributed to the secular increase in cash holdings. Consistent with Duchin (2010), we show that the increase in the value of cash coincides with the decrease in diversification in multi-segment firms. Finally,

[5] As noted earlier, the decade of the 1990s witnessed a far greater increase in the average marginal value of cash than the 2000s, which can explain why many of the results documented here are statistically and economically more pronounced during the decade of the 1990s.

[6] Cooper and Jensen (2016) document that the value premium to cash holdings increases with short-term borrowing cost as proxied by the federal funds rate, the TED spread, and spreads in the commercial paper market. Similarly, Harford, Klasa, and Maxwell (2014) find that the marginal value of cash is increasing in the prevailing commercial and industrial (C&I) rate, particularly for firms with relatively short-term debt maturities.

investor sentiment can affect the perceived value of future investment opportunities and by extension the precautionary value of cash. We find that investor sentiment can explain a small fraction of the increase in cash value from the 1980s to the 1990s.

After controlling for the time-varying effects of these economic forces, the unexplained portion of the rise in the value of cash is statistically negligible over the sample period. While these results are economically suggestive, we acknowledge that we cannot draw definitive causal conclusions from our analyses in the absence of an exogenous shock that spans our multi-decade sample horizon.

The persistent cash deficits documented by Bates et al. (2009) are consistent with our finding that the marginal value of cash has increased despite a coincident surge in cash holdings during the sample period. One plausible explanation for this phenomenon is that market frictions prevent firms from quickly adjusting to their optimal cash levels. To identify the impact of financing frictions, we implement a partial adjustment model and estimate the speed of adjustment (SOA) to the optimal level of cash over time. Our results indicate that the SOA has declined over successive decades, and significantly more so for firms with cash deficits. Furthermore, we find that the slowdown in the SOA for cash-deficit firms is more pronounced for firms facing greater financial constraints.

Finally, we examine whether agency costs can explain the secular variation in the value of cash. The literature has established a strong cross-sectional relation between the quality of corporate governance and the marginal value of cash. For example, Dittmar and Mahrt-Smith (2007) observe that the value of cash is higher for firms that are better governed as measured by both the presence of institutional blockholders and the Gompers, Ishii, and Metrick (2003) G-index of shareholder rights. Using the Cremers and Nair (2005) measure of cumulative institutional ownership, we find that institutional blockholdings are uncorrelated with the marginal value of cash in the 1980s, but can explain a small fraction of the increase for the 1990s. While we confirm that the G-index is correlated with the value of cash in the cross section of firms, its effect on the value of cash does not vary over time.

II. Sample Selection and Data Description

We obtain our sample of firm-year observations from the Center for Research in Security Prices (CRSP)/Compustat merged database for the years 1980 through 2009. These observations consist of surviving and non-surviving firms that appear in Compustat at any time during the sample period. We exclude financial firms (Standard Industrial Classification (SIC) codes 6000–6999) because these firms hold cash to maintain reserve requirements. We also exclude utilities (SIC codes 4900–4999) because these firms are subject to regulatory oversight. We require that a firm-year observation has positive total assets (Compustat annual item AT) to be included in our sample. Our final sample consists of 12,845 unique firms and 108,357 firm-year observations.

To estimate the marginal value of cash, we follow the methodology described in Faulkender and Wang (2006).[7] Specifically, we compute a firm's excess equity return as its nominal return during a fiscal year, minus the benchmark portfolio return estimated using the Fama–French (1993) 25 size and book-to-market portfolios. We then estimate the determinants of excess annual returns as follows:

$$(1)\quad r_{i,t} - R^B_{i,t} = \gamma_0 + \gamma_1 \frac{\Delta C_{i,t}}{M_{i,t-1}} + \gamma_2 \frac{\Delta E_{i,t}}{M_{i,t-1}} + \gamma_3 \frac{\Delta \mathrm{NA}_{i,t}}{M_{i,t-1}} + \gamma_4 \frac{\Delta \mathrm{RD}_{i,t}}{M_{i,t-1}} + \gamma_5 \frac{\Delta I_{i,t}}{M_{i,t-1}} + \gamma_6 \frac{\Delta D_{i,t}}{M_{i,t-1}}$$
$$+ \gamma_7 \frac{C_{i,t-1}}{M_{i,t-1}} + \gamma_8 L_{i,t} + \gamma_9 \frac{\mathrm{NF}_{i,t}}{M_{i,t-1}} + \varepsilon_{i,t}\ ,$$

where $r_{i,t} - R^B_{i,t}$ is the excess stock return for firm *i* during fiscal year *t* and the Δ terms identify unexpected changes in explanatory variables including cash holdings (*C*), earnings (*E*), net assets (NA), R&D (RD), interest expense (*I*), dividends (*D*), financial leverage (*L*), and net financing (NF). Since both the dependent and the independent variables are standardized by lagged market

[7] The model in equation (1) has been used in a number of studies to examine cross-sectional variation in the value of cash including Dittmar and Mahrt-Smith (2007), Denis and Sibilkov (2010), Liu and Mauer (2011), Harford et al. (2014), and Chen et al. (2015). The specification is based on prior work in Fama and French (1998), Pinkowitz and Williamson (2004), and Pinkowitz et al. (2006).

value, the coefficient γ_1 is an estimate of the marginal value of one additional dollar of cash holdings.

The explanatory variables in the model control for characteristics that are correlated with investor's expectations about fundamentals that affect firm value. If the changes in value-relevant firm characteristics are adequately accounted for, the coefficient estimate on the change in cash can be interpreted as the marginal value of cash. It is possible that an omitted correlated variable may cause the excess return and the change in cash to be endogenously determined, although to date no study has identified such a variable. Nevertheless, we acknowledge the endogeneity concern and address it when we discuss our empirical findings.

Our empirical approach largely follows Opler, Pinkowitz, Stulz, and Williamson (1999) and Bates et al. (2009) in considering the following as potential determinants of the secular variation in the value of cash:

R&D (RD). We set missing R&D values to 0. In the regressions we construct a dummy variable for R&D as either low or high, classified using the annual median R&D-to-assets ratio for firms with nonmissing and nonzero R&D.

Market-to-book ratio ((AT – CEQ + (PRCC_F × CSHO)) / AT). In our specifications we use the logarithm of the market-to-book ratio (MB) because of its right skewness.

Cash-flow volatility (CASH-FLOW_VOLATILITY). Following Opler et al. (1999), we estimate this at the industry level. For each firm in a given year we compute the standard deviation of its cash-flow-to-net-assets ratio ((OIBDP – XINT – TXT – DVC) / (AT – CHE)) using the last 20 years of data, requiring at least five annual observations. For each year, we then take the average cash-flow standard deviation for each 2-digit SIC code, and use this industry average as the measure of cash-flow volatility.

SA (size and age) financial constraint index (SA_INDEX). We compute the SA index following Hadlock and Pierce (2010). The index is constructed as: $(-\ 0.737 \times \text{Size}) + (0.043) \times \text{Size}^2 - (0.040 \times \text{Age})$. Higher values of the index are associated with greater financial constraints.

Credit rating dummy (CREDIT_RATING_DUMMY). We construct a credit rating dummy similar to Almeida, Campello, and Weisbach (2004). The dummy equals 1 if a firm's long-term credit rating is junk grade or not available, and 0 if the credit rating is investment grade.

Credit spread (CREDIT_SPREAD). We obtain monthly corporate bond yield information from the Federal Reserve Bank of St. Louis and compute the spread as the yield on Aaa bonds minus the yield on Baa bonds.[8] We then compute the average 12-month spread corresponding to each firm's fiscal year.

IPO dummy (IPO_DUMMY). We obtain the year of a firm's initial public offering (IPO) from Thomson's SDC database. If unavailable, we assume it is the year prior to the first year that the firm appears in Compustat. The IPO dummy is equal to 1 if the firm went public within the prior 5 years.

Institutional block holdings (5%_BLOCK). We obtain institutional ownership from the Thomson Reuters Institutional Holdings (13F) Database. Following Cremers and Nair (2005), we aggregate institutional holdings of more than 5% of a firm's equity.

Herfindahl–Hirschman Index (HHI). HHI index is a concentration measure computed for each industry-year as the sum of squared market share of each firm in each Fama–French 48 industry.

Multi-segment dummy (MULTISEG_DUMMY). The multi-segment dummy is an indicator variable equal to 1 if a firm reports three or more business segments in a given year, and 0 otherwise.

Investor sentiment (SENTIMENT). Following Baker, Wurgler, and Yuan (2012) we compute four annual proxies for investor sentiment: i) the volatility premium as the log of the ratio of the value-weighted average market-to-book ratio of high-volatility stocks to that of low-volatility stocks; ii) total volume of IPOs; iii) first-day average return of IPOs; and

[8] http://research.stlouisfed.org/fred2/categories/119.

iv) market turnover. We then orthogonalize each of the four proxies to six macroeconomic variables: consumption growth, industrial production growth, inflation, employment growth, the short-term rate, and the term premium. Finally, we extract the first principal component from the four orthogonalized sentiment proxies as our composite measure of investor sentiment. Since sentiment is measured for each calendar year, we match sentiment to the closest fiscal year-end for each observation.

Table 1 provides summary statistics for the variables described above. We also incorporate a number of other firm-level characteristics in our analyses. Cash equals cash plus marketable securities (CHE). Net assets are measured by total assets minus cash (AT – CHE). Firm size is measured as the logarithm of the book value of assets (AT) adjusted to 2004 dollars. The market value of equity is calculated as the per-share stock price times number of shares outstanding (PRCC_F × CSHO). Earnings are computed as earnings before extraordinary items plus interest, deferred tax credits, and investment tax credits (IB + XINT + TXDI + ITCI). Interest expense is Compustat item (XINT). Net financing is total stock issuance minus stock repurchase (SSTK – PRSTKC) plus debt issuance minus debt redemption (DLTIS – DLTR). Market leverage is estimated as long-term debt plus short-term debt divided by the sum of long-term debt, short-term debt, and the market value of equity ((DLTT + DLC) / ((DLTT + DLC) + (PRCC_F × CSHO))). We winsorize all continuous variables at the 1st and 99th percentiles.

[Insert Table 1 about here]

III. The Secular Increase in the Value of Cash and Excess Cash

In Table 2 we report the annual estimates of the marginal value of cash from 1980 to 2009, and the 5- and 10-year moving averages. The 10-year moving average indicates that an additional dollar of cash holdings is valued at $0.61 in the 1980s, $1.04 in the 1990s, and $1.12

in the 2000s.[9] The correlation between time and the value of cash is 0.59. In a regression of the annual estimates of the value of cash on a constant and time, the adjusted R^2 is 32%, and time has a positive and significant coefficient that implies an average annual increase of 2.5 cents in the marginal value of one dollar of cash. In Figure 1, we graph the annual estimates of the value of cash and the 5- and 10-year moving averages. The marginal value of cash peaks in 1999, a period marked by high market valuation for growth opportunities. If we exclude the year 1999, the correlation between time and the value of cash increases to 0.64, and in a regression of annual value of cash estimates on a constant and time, the adjusted R^2 increases to 44%.[10] Overall, the increase in the value of cash holdings coincides with the previously documented secular increase in the level of cash holdings.

[Insert Table 2 about here]

[Insert Figure 1 about here]

To evaluate the robustness of the findings presented in Table 2, we employ an alternative valuation model introduced in Pinkowitz and Williamson (2004) and Pinkowitz, Stulz, and Williamson (2006), which is a variation of the model in Fama and French (1998). The model uses the market-to-book-assets ratio as the dependent variable.[11] When we employ the Pinkowitz et al. (2006) methodology, we find a similar secular increase in the value of cash. The time-series correlation between the value-of-cash estimates from the two methods is 0.77. In Figure 1, we also plot the 10-year moving average of the value of cash using the Pinkowitz et al. (2006) method and it resembles the 10-year moving average obtained using the Faulkender and Wang (2006) method.

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[9] We report our results for distinct decades following convention in the literature. This delineation is largely ad hoc so we also consider differences over more discrete subperiods later in the paper.

[10] To check whether our findings are unduly influenced by extreme values in excess returns, we winsorize excess return at the 5th and 95th percentiles. The findings in Table 2 and all the following tests remain qualitatively similar with this approach.

[11] Faulkender and Wang (2006) argue for using excess stock return as the dependent variable because it incorporates time-varying risk factors into the valuation measure and because returns are more straightforward to measure and interpret than the market-to-book ratio.

The results in Bates et al. (2009) suggest that during the sample period, the average U.S. industrial firm held less cash than predicted by models of the transaction and precautionary demand for cash. Thus, a natural question is whether the trend in the marginal value of cash can be explained by trends in cash deficits. To the extent that a liquidity deficit exacerbates the underinvestment problem, we predict that cash deficits will be associated with a higher marginal value of cash. Alternatively, a surplus of cash increases the agency costs of managerial discretion, leading to a lower marginal value. To estimate excess cash, we start from a regression model that relates the cash ratio to various firm-level determinants following Opler et al. (1999). We then use the coefficients from this model to decompose cash into excess cash and predicted cash. Specifically, we run Fama–MacBeth (1973) regressions of the cash-to-total-assets ratio on the determinants of the level of cash over the 30-year sample period, and use the average coefficients to predict the level of cash. For each firm-year observation, excess cash is the difference between actual and predicted cash holdings.

In each sample year, we sort firms into terciles of excess cash, retain the top and the bottom terciles, and use equation (1) to estimate the value of cash for the top and bottom terciles of firms.[12] In keeping with our prediction, we find that the value of cash is significantly higher for firms with negative excess cash (the bottom tercile) than for those with positive excess cash (the top tercile). Figure 2 plots the 10-year moving average of the value of cash for these two groups. During the sample period, the increase in value of cash is more pronounced for firms with negative excess cash than for those with positive excess cash, particularly since the early 1990s. In Section VI of the paper we address differences in the speed of adjustment in cash holdings for firms with positive and negative excess cash.

[Insert Figure 2 about here]

[12] Specifically, we run an augmented version of equation (1) annually: $r_{i,t} - R^B_{i,t} = \gamma_0 + \gamma_{1a}\,\mathrm{NEGEX} \times \frac{\Delta C_{i,t}}{M_{i,t-1}} + \gamma_{1b}\,\frac{\Delta C_{i,t}}{M_{i,t-1}} + \gamma_2\,\mathrm{NEGEX} + \gamma_3\,\frac{\Delta E_{i,t}}{M_{i,t-1}} + \gamma_4\,\frac{\Delta \mathrm{NA}_{i,t}}{M_{i,t-1}} + \gamma_5\,\frac{\Delta \mathrm{RD}_{i,t}}{M_{i,t-1}} + \gamma_6\,\frac{\Delta I_{i,t}}{M_{i,t-1}} + \gamma_7\,\frac{\Delta D_{i,t}}{M_{i,t-1}} + \gamma_8\,\frac{C_{i,t-1}}{M_{i,t-1}} + \gamma_9\,L_{i,t} + \gamma_{10}\,\frac{\mathrm{NF}_{i,t}}{M_{i,t-1}} + \varepsilon_{i,t}$, where NEGEX is a dummy variable equal to 1 if the firm-year observation is in the bottom-tercile excess-cash group. Thus, γ_{1b} is the value of cash for the top-tercile excess-cash group, and ($\gamma_{1b}+\gamma_{1a}$) is the value of cash for the bottom-tercile excess-cash group.

IV. How Pervasive is the Secular Increase in the Value of Cash?

A. Subsample Analysis

We first consider the prevalence of the increase in the value of cash by examining the trend for various subsamples of firms. We begin by sorting our sample firms annually into quintiles based on their book value of assets, and plot the 10-year moving average of value of cash for each size quintile in Figure 3. Small firms have a higher marginal value of cash than large firms, a finding consistent with economies of scale in the transactions demand for cash holdings (see, e.g., Mulligan (1997)). It seems unlikely, however, that the transaction motive can fully explain the secular increase in the value of cash because the value of cash increases for both large and small firms. Regression point estimates indicate that for firms in the smallest size quintile, the value of one additional dollar of cash for the 1980s, the 1990s, and the 2000s is $0.69, $1.07, and $1.11, respectively. For the largest size quintile, the estimates are $0.39, $0.66, and $0.72.

[Insert Figure 3 about here]

The precautionary demand for cash suggests that firms hold cash to cope with adverse shocks to investment when access to external capital is costly (e.g., Almeida et al. (2004)). Opler et al. (1999) also note that the precautionary motive suggests that firms with better investment opportunities will hold more cash because adverse cash flow shocks and financial distress are more costly for them. To evaluate the importance of investment opportunities on the secular variation in value of cash, we first use a firm's R&D expense as a proxy for investment opportunities. In Figure 4 we outline the value of cash for three groups of firms: those with above or below annual sample median R&D expenditures, and those with missing or zero R&D. In the 1990s, the marginal value of cash increases for all three subsamples, but the increase is most pronounced for the subsample of high R&D firms. This evidence is consistent with the finding in He and Wintoki (2016) that the demand for cash increased over time for R&D intensive firms. In the 2000s, the value of cash for firms with low and no R&D continues to increase while the value of cash for high R&D firms declines slightly.

[Insert Figure 4 about here]

Following Loughran and Ritter (2004), we use a technology industry classification as an alternative proxy for investment opportunities. As documented in Figure 5, both technology and non-technology firms experience an increase in the value of their cash holdings. Technology firms experience a slight drop in the value of cash over the 2000s, while non-technology firms continue to see the value of cash rise in the 2000s.

[Insert Figure 5 about here]

In Figure 6, we use the market-to-book ratio (MB) as an additional proxy for investment opportunities, and sort firms annually into MB quintiles. The increase in value of cash is most notable for high MB firms in the 1990s. During the 2000s, the value of cash stays relatively flat for high MB firms, but keeps increasing for low MB firms. In sum, Figures 4, 5, and 6 suggest that investment opportunities were an important driver of the secular increase in the value of cash, particularly for firms with high growth opportunities in the 1990s. During the 2000s, the value of cash continues to increase for firms with lower growth opportunities.

[Insert Figure 6 about here]

Under the precautionary motive, cash holdings are more valuable when operating cash flows are more volatile. To identify whether the secular trend in the value of cash holdings is related to cash-flow volatility, we sort firms annually into volatility quintiles. As summarized in Figure 7, the value of cash increases for all quintiles over the sample period, and is most noticeable for higher quintiles during the 1990s.

[Insert Figure 7 about here]

Financial constraints also increase a firm's precautionary motive for holding cash and, all else equal, the marginal value of cash. Small firms are likely more financially constrained than large firms, and as shown in Figure 3, smaller firms indeed exhibit higher valuation of cash holdings. We also follow Hadlock and Pierce (2010) and measure financial constraints using both firm size and age. We sort our firm-year observations annually into quintiles formed on the size-and-age (SA) index. As seen in Figure 8, the marginal value of cash exhibits an uptrend across all quintiles, and as expected, the value of cash is consistently higher for more constrained firms. The rate of increase in the value of cash falls in the 2000s, particularly for more constrained firms, but continues to rise for firms that are the least constrained. While these results seem puzzling, they may be due to confounding effects in the supply of alternative sources of liquidity or changes in other firm features correlated with the SA index. We control for these effects in multivariate models that follow.

[Insert Figure 8 about here]

We also use a firm's bond credit rating as a proxy for financial constraint. Following Almeida et al. (2004), we classify firms as financially unconstrained if they have investment grade credit ratings, and as financially constrained if they have junk or no crediting ratings. Figure 9 shows that the upward trend in the value of cash is present for both groups, particularly in the 1990s.

[Insert Figure 9 about here]

Foley, Fritz, Hartzell, and Twite (2007) find that cash holdings are increasing in the tax consequences of repatriating foreign cash reserves. While maintaining foreign reserves minimizes the tax consequences associated with repatriation of cash for domestic investment or distribution to investors, the value of these reserves is limited by the precautionary demand within a particular jurisdiction. Blouin and Krull (2009) observe that firms with relatively poor investment opportunities were more likely to repatriate cash under the American Jobs Creation Act of 2004. While foreign cash reserves might effectively be "trapped" overseas, leading to discounted valuations, it is also possible that the buildup of foreign cash reserves is positively correlated with future growth opportunities in a particular country, increasing the marginal value of foreign held cash over time.

Since most U.S. corporations do not report the specific locations where cash reserves are held, we consider the proportion of pre-tax income generated in foreign countries. We examine the value of cash for companies delineated into three subgroups: those with i) no foreign income, ii) below-median foreign income, or iii) above-median foreign income. The median pre-tax foreign-income-to-total-assets ratio is obtained annually using all firms reporting non-zero foreign income. Data on foreign income are available in Compustat beginning in 1980, but initial reporting is sporadic until 1985, which is when we begin to collect the data for this analysis.

We estimate the marginal value of cash for the three foreign income subgroups each year, and plot the 10-year moving average in Figure 10. There is no obvious difference in the value of cash between the three subgroups, and all exhibit an upward trend in the marginal value of cash up to the early 2000s. The results suggest that on average, foreign cash reserves are not an important determinant of the secular trend in the value of cash.

[Insert Figure 10 about here]

Bates et al. (2009) find that the increase in cash holdings is partly explained by a change in the composition of publicly held firms over time. The marginal value of cash held by recent IPO firms is usually higher because these firms tend to have more valuable investment opportunities and greater cash-flow volatility. Figure 11 plots the value of cash over time for IPO cohorts sorted by 5-year IPO subperiods. The marginal value of cash for firms of more recent IPO vintage is indeed higher. We also observe secular differences in the value of cash within the IPO cohorts. For example, IPO cohorts in the 1990s and 2000s see the value of cash decline during the 2000s, while firms that went public in the 1960s and 1970s actually see their value of cash rise in the 2000s.

[Insert Figure 11 about here]

To further consider the effect of firm age, we estimate the value of cash for the constant composition subsample of 400 public firms that survive for the entire 30-year sample period. This subsample accounts for approximately 10% of the overall firm-year observations. The trend in the marginal value of cash for the constant composition subsample is similar to that observed for the broader sample: $0.71 in the 1980s, $0.84 in the 90s, and $1.00 in the 2000s.

We examine the relation between a number of other individual firm features and the trend in the value of cash, but avoid additional illustration for brevity. In general, we find that the increase in the marginal value of cash over time is higher for firms with greater capital expenditures, net working capital, and acquisition expenses. While the analyses in this section suggests that the increase in the marginal value of cash is quite pervasive, we conduct more formal multivariate analyses to identify the drivers of the secular increase in the value of cash in the next section of the paper.

B. Multivariate Analysis

In Table 3 we summarize the estimates of the marginal value of cash over time utilizing the framework in equation (1). As described in Section II, we regress a firm's excess equity return on the change in cash holdings and a set of control variables. In all regressions we report *t*-statistics that are robust to heteroscedasticity and firm and year clustering. We employ time indicator variables for the decades of the 1990s and the 2000s. We utilize decade indicators because it allows us to maintain a relatively parsimonious regression model and is consistent with analyses of time trends in the level of cash holdings in the prior literature.[13]

[Insert Table 3 about here]

Model 1 of Table 3 excludes the decade indicator variables. In this specification, the marginal value of one additional dollar of cash is $1.078 during the full sample period. As we note in Section II, it is possible that the coefficient estimate for the change in cash does not capture the marginal value of cash, but rather the effect of some omitted firm characteristic. For example, the change in cash may be partially related to future profitability, which is reflected in excess stock returns. We note, however, that the empirical model controls for the contemporaneous change in profitability through the change in earnings. To test whether change in cash predicts future excess returns, we incorporate three lags of the change in cash in a model otherwise identical to model 1 of Table 3. In this untabulated specification, none of the coefficients associated with lagged changes in cash are statistically significant.

[13] For robustness, we perform regressions identical to those in Table 3, but for 5-year subperiods from 1980 to 2009. The trends over these shorter subperiods comport with the broader trends summarized in Table 3.

In model 2 of Table 3, we introduce decade indicators for the 1990s and the 2000s. The point estimates for the interaction terms between the change in cash and decade indicators relay the incremental value of cash during these decades relative to the 1980s. The results in model 2 indicate that in the 1980s, a $1 increase in cash holdings is valued at $0.711. In the 1990s, the marginal value of cash increases by $0.498 to $1.209 (= $0.711 + $0.498). In the 2000s, the value of an incremental dollar continues to increase, albeit at a much slower pace, to $1.263 (= $0.711 + $0.552).

Faulkender and Wang (2006) find that the marginal value of cash decreases in the level of cash and financial leverage. In model 3 of Table 3, we control for cash and leverage effects over time by incorporating three-way interaction terms between the change in cash, the decade indicators, and the level of cash and financial leverage. After controlling for the level of cash and financial leverage, the value of $1 in additional cash is $0.821 in the 1980s where the sample means of the level of cash and financial leverage are 0.176 and 0.234, respectively.[14] Relative to the 1980s, $1 in additional cash is worth $0.505 more in the 1990s and $0.721 more in the 2000s.[15] Thus, even after controlling for temporal changes in the level of cash and financial leverage, the increase in the value of cash remains economically significant.

A second, more interesting result in model 3 of Table 3, is that the negative value effects associated with the level of cash and leverage have become more pronounced over time, providing some insight into the firm features behind the trend in value. Faulkender and Wang (2006) posit that taxes on capital gains and dividends lead to discounts in the marginal value of cash for firms with higher levels of cash. Our results are consistent with the time trend in the average effective long-term capital gains rate, which increased from a low of 14.5% in 1982 to a high of 25.1% in 1996.[16] Similarly, dividend tax rates increased in 1985 to the individual

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[14] Specifically, 0.821 = 1.092 – 0.262 × 0.176 – 0.962 × 0.234. If we drop the –0.262 × 0.176 term because the coefficient estimate on $\Delta C_t \times C_{t-1}$ is not statistically significant at the 10% level, 0.821 becomes 0.867.
[15] Specifically, 0.505 = 0.931 – 0.581 × 0.176 – 1.382 × 0.234; 0.721 = 1.045 – 0.636× 0.176 – 0.905 × 0.234.
[16] Source: http://www.treasury.gov/resource-center/tax-policy/Documents/OTP-CG-Taxes-Paid-Pos-CG-1954-2009-6-2012.pdf.

income tax rate, which varied from 28% to 50% during the sample period. The dividend rate was subsequently lowered to 15% in 2003.

The effect of leverage on the value of cash also becomes more pronounced over time, a result consistent with a coincident decrease in average debt maturity. Faulkender and Wang (2006) suggest that if firms utilize cash to pay down debt, a fraction of the value associated with incremental cash savings will accrue to the firm's debt holders. Custódio, Ferreira, and Laureano (2013) and Harford, Klasa, and Maxwell (2014) find that the maturity of debt held by U.S. corporations dropped precipitously over the last 30 years, leading to an increase in interest rate and refinancing risk. Therefore, for a given level of debt, the value of cash accrues less to shareholders and more to debt holders to compensate for this increasing risk. Harford et al. (2014) indicate that the decrease in debt maturity increased the demand for cash holdings in firms with relatively more debt.

C. Robustness and Endogeneity Concerns

As documented in the second column of Table 2, annual estimates of the marginal value of cash are noisy, with the years 1999 and 2003 exhibiting the highest values during the sample period. Coincidentally, these 2 years also yield large market returns where the respective annual Standard & Poor's (S&P) 500 Index returns are 21.0% and 28.7%. It is possible that the coefficient estimates are driven by the dependent variable rather than the change in the marginal value of cash. This explanation is unlikely for several reasons. First, the excess return is a market adjusted return.[17] Second, the correlation between the annual value-of-cash estimates and the S&P 500 annual returns is quite low (0.146). Third, if we exclude the years 1999 and 2003 from our regressions, we obtain qualitatively similar results to those in Table 3, and the interaction terms between the change in cash and the two decade indicators remain economically

[17] The dependent variable in equation (1), excess return, is adjusted by the benchmark size-and-book-to-market portfolio return. For robustness, we alternatively adjust the stock return by the risk-free rate, and include the 3 Fama–French (1993) factors and momentum as control variables. The findings in Table 3 remain qualitatively similar.

large and statistically significant. Finally, if the increase in the value-of-cash estimate is driven by the dependent variable, we should observe an increase in the estimates of other regressors, for example ΔRD and financial leverage, but we do not.

Another potential alternative explanation for our results in Table 3 is that the change in cash value relates to future profitability, and firms may have become more profitable over time. As we noted earlier, our valuation model controls for the change in earnings, which should capture a significant portion of anticipated future profitability. Nonetheless, we address this concern in more detail. First, in untabulated regressions we incorporate three years of forward-looking changes in earnings (ΔE) in a model otherwise similar to model 1 of Table 3. If excess returns reflect unmodeled future profitability, the coefficient estimates for forward ΔE should be positive and significant. Our results indicate that the coefficient estimate for forward 1 year ΔE is only marginally positive and statistically significant (coeff. = 0.10, *t*-stat. = 1.73), while the coefficients for forward 2- and 3-year ΔE are not statistically different from 0. Given this evidence, we conclude that contemporaneous ΔE effectively captures the profitability component of excess returns.

A related concern is that the increase in the value of cash over time might be attributable to profitable firms becoming more profitable over time, leading to higher excess returns. To consider this possibility, we rerun the specifications in Table 3 and incorporate interaction terms between the change in earnings (ΔE) and the decade indicator variables. If the increase in the value of cash is a function of profitable firms becoming more profitable over time, the interactions between ΔE and the decade indicators should subsume the significance of the interaction terms between ΔC and the decade indicators. We do not find this to be the case. For example, in an untabulated specification otherwise similar to model 3 of Table 3, the coefficient estimates for the interactions between ΔC and the decade variables drop only slightly and retain their statistical and economic significance. Specifically, $\Delta C \times$ DUM_1990 is 0.924 (*t*-stat. = 2.58), and $\Delta C \times$ DUM_2000 is 1.005 (*t*-stat. = 2.72). In the same regression, the coefficient estimate for $\Delta E \times$ DUM_1990 is 0.13 (t-stat = 2.22) and the coefficient for $\Delta E \times$ DUM_2000 is not significantly different from zero (t-stat. = 1.12).[18]

V. Why Has the Value of Cash Increased Over Time?

A. Theory and Empirical Hypotheses

The results in Section IV indicate that public firms in the U.S. experienced a significant secular increase in the value of their cash holdings between 1980 and the end of the 2000s. This increase was both statistically and economically significant, with the most pronounced effects evidenced during the 1990s where, on average, $1 of incremental cash was worth approximately 70% more than in the 1980s. The evidence is quite robust to controlling for a variety of cross-sectional differences in firm characteristics and does not appear to be the byproduct of model misspecification or spurious correlation. Our findings closely parallel the well-documented increase in the level of cash holdings over the same time period. While this may seem counterintuitive given that the marginal value of cash is expected to decline as firms add to their liquidity stock, we note that Bates et al. (2009) find that the increase in the level of cash holdings was largely determined by changes in firm fundamentals, which in turn likely drives the valuation effects that we observe in our models.

We draw on existing theoretical and empirical studies of the determinants of cash holdings to develop testable hypotheses for why the value of cash increased over time. Keynes (1936) describes two motives for holding cash. Under the transaction motive, cash is valuable because firms incur transactions costs associated with converting noncash financial assets into cash for payments. Under the precautionary motive, corporations hold cash to hedge negative cash flow shocks and avoid costly securities issuance. Theoretical work, such as Gamba and Triantis (2008) and Riddick and Whited (2009), builds on this intuition. The empirical findings

[18] We also sort firms each year into ROA quintiles, and estimate the value of cash for each quintile over time. If the increase in the value of cash is driven by more profitable firms, we expect to see the increase in the value of cash to be more significant for high ROA firms. We do not observe this. Rather, the increase in the value of cash is similar across ROA quintiles.

in the literature generally support the transaction motive for cash with scale economies. The precautionary demand for cash, however, is a more significant determinant of observed cash holdings. For example, Kim, Mauer, and Sherman (1998) and Opler et al. (1999) find that cash holdings are positively correlated with a firm's growth opportunities, cash-flow volatility, and financial constraints. In addressing the determinants of the secular variation in the level of cash holdings, Bates et al. (2009) show that the increase during our sample period can be largely attributed to changes in these characteristics for existing firms, as well as an influx of risky and financially constrained IPO firms. Given the increasing precautionary demand for cash holdings over time, we expect that these same characteristics will give rise to a positive trend in the marginal value of cash as firms attempt to adjust liquidity in the face of changing business conditions.

While the increase in the marginal value of cash is persistent, its relation to any single characteristic is likely to be time varying. As noted in Bates et al. (2009), firms adjust their levels of cash holdings in response to changing firm characteristics over time. For example, they show that an increase in cash flow risk accounted for much of the increase in cash holdings, particularly during the 1990s, and less so in the 2000s. Given some adjustment costs, firms will respond to changes in these conditions by increasing or decreasing their cash holdings to the new optimum. Since aggregate effects tied to the demand for cash are positive over the three decades of our study, we infer that these adjustments are, on average, positive and associated with an increase in the marginal value of cash.

Given the expected relation between precautionary motives for holding cash and its marginal value for the firm, we outline the following two hypotheses in alternative form:

Hypothesis 1. The increase in the value of cash over time can be partially explained by the influx of IPO firms during the sample period, and the commensurate change in the composition of the U.S. public market.

Hypothesis 2. Characteristics that increase (decrease) the precautionary motive for cash holdings also increase (decrease) the marginal value of cash holdings.

While most of the 1980s and the 1990s witnessed relative economic prosperity, the 2000s were characterized by two economic recessions. In particular, heightened macroeconomic and capital market uncertainty during the recession in the late 2000s could have contributed to the continued increase in both the demand for, and the value of, corporate cash holdings. Consistent with this, Duchin et al. (2010) find that the credit market supply shock associated with the financial crisis had the greatest effect on investment for firms with low cash reserves and firms that are largely dependent on external capital. Thus we propose a third hypothesis:

Hypothesis 3. Heightened macroeconomic and capital market uncertainty in the 2000s contributed to the increase in the value of cash holdings during this decade.

We also consider how the previously documented increase in product market competition influenced the value of corporate cash holdings during our sample period. Irvine and Pontiff (2009) show that this trend in competition contributed to greater volatility of operating cash flows and stock returns. Research by Frésard (2010), Schroth and Szalay (2010), Morellec, Nikolov, and Zucchi (2014), and Lyandres and Palazzo (2016) suggests that cash provides firms with a valuable strategic advantage in competitive product markets. Consistent with this, Chi and Su (2016) and Alimov (2014) find that in the cross section, more intense competition leads to a higher value of cash. Given these findings we propose the following hypothesis:

Hypothesis 4. The increase in product market competition contributed to the positive trend in the marginal value of cash over the sample period.

Duchin (2010) documents that the degree of cross-segment diversification in multi-segment firms decreased from the 1980s, resulting in a higher precautionary demand for cash. We expect that the secular decrease in diversification for multi-segment firms during the sample period also contributed to the increase in the marginal value of cash, leading to the following hypothesis:

Hypothesis 5. The decrease in diversification in multi-segment firms increases the value of cash holdings.

Investor sentiment can affect the perceived value of future investment opportunities and, by extension, the value of precautionary cash holdings. The 1990s witnessed particularly high investor sentiment, which reversed significantly at the burst of the tech bubble in the early 2000s. Given this evidence, we propose the following hypothesis:

Hypothesis 6. Time-varying investor sentiment can explain the trend in the marginal value of cash over time.

B. The Impact of Individual Characteristics on the Value of Cash Over Time

In Table 4 we supplement the regressions outlined in Table 3 with additional terms including the interaction between the change in cash with the two decade indicators, and firm and macroeconomic characteristics that can potentially explain the change in the value of cash over time. In Panel A of Table 4 we examine each potential determinant separately. In Panel B we examine all the potential determinants simultaneously.

We introduce the IPO cohort dummy in model 1 of Panel A. If cash-flow volatility and growth opportunities are higher for firms that IPO later in the sample period, the marginal value of cash should be higher for firms in later IPO cohorts. Consistent with Hypothesis 1, the results of model 1 indicate that the marginal value of cash is higher for firms in successively later IPO cohorts. Compared with the 1980s, the value of cash for IPO firms in the 1990s is \$0.329 higher, a result that coheres with the observation that this decade was characterized by a surge in IPO activity by technology firms. The change in the value of cash for IPOs in the 2000s is not statistically significant. When comparing the estimates on $\Delta C_t \times$ DUM_1990 and $\Delta C_t \times$ DUM_2000 with those from model 3 of Table 3, we observe that newly issued IPO firms can explain part of the increase in the value of cash. The coefficient estimate on $\Delta C_t \times$ DUM_1990 drops by 12% from \$0.931 to \$0.821 and $\Delta C_t \times$ DUM_2000 drops by 2% from \$1.045 to \$1.023.

[Insert Table 4 about here]

In models 2–4 of Panel A in Table 4 we examine the implications of Hypothesis 2. We begin by modeling the effects of investment opportunities over time in model 2. As described in Section II, we construct an indicator variable for R&D expenditures where firm-years with no, or below-median, R&D expenditures take on the value of 0, and above-median R&D expenditures take on the value of 1. Similar to what we observe in Figure 4, the marginal value of cash associated with R&D expenditures increases from the 1980s to the 1990s, but decreases somewhat in the 2000s. While still significant, the coefficient estimate for the interaction term between the change in cash and the 1990s indicator declines relative to model 1 (from 0.821 to 0.572), suggesting that during the 1990s firms with significant R&D expenditures accounted for an increase in the marginal value of cash even after controlling for IPO cohort effects. Controlling for the valuation consequences of R&D also helps to explain why IPO firms in the 1990s experienced an increase in their value of cash as the coefficient estimate for $\Delta C_t \times$ IPO_DUMMY × DUM_1990 becomes insignificant in model 2. The decline in the marginal value of cash for firms with relatively high R&D expenditures in the 2000s is consistent with the decline of the technology bubble and the coincident reduction in aggregate market capitalization tied to technology firms. We note that this coefficient does not indicate that R&D has a negative effect on the value of cash in the 2000s, but just that the effect is muted relative to the 1990s. For robustness, we also use the market-to-book ratio (MB) in lieu of R&D expenditures in a regression otherwise identical to model 2 and find qualitatively similar results.

In model 3 of Panel A in Table 4 we examine whether cash-flow volatility can explain the increase in the value of cash. For the 1980s and the 2000s, the marginal value of cash does not vary with cash-flow volatility. For the 1990s, however, a 1-standard-deviation increase in cash-flow volatility is associated with $0.49 increase in the marginal value of cash. Cash-flow volatility also explains a significant portion of the overall increase in the value of cash during the 1990s where the coefficient on $\Delta C_t \times$ DUM_1990 is now statistically insignificant. The increase in the marginal value of cash in the 2000s remains significant, where the coefficient on $\Delta C_t \times$ DUM_2000 = 0.776 (*t*-stat. = 2.03).

In model 4 of Panel A in Table 4 we examine whether financial constraints, measured by quintiles of the SA index, can explain the increase in the value of cash. In this regression, we construct the SA quintiles by decade. The results suggest that the marginal value of cash is higher for financially constrained firms, and the effect associated with financial constraints increases from the 1980s to later decades. In untabulated tests, we use four alternative measures of financial constraints that include quintiles of the Whited–Wu (2006) index, whether a firm has bond credit rating, whether a firm has commercial paper credit rating, and whether a firm is a dividend payer. We find that each of these alternative measures has a positive and significant impact on the marginal value of cash for the cross section of firms, and that the marginal value of cash is increasing in each measure of financial constraint from the 1980s through the decades of the 1990s and 2000s.

Given that the U.S. economy suffered two recessions during the decade of the 2000s, we extend our model to include credit spread as a factor to explain the increase in the marginal value of cash. Credit spread, measured by the Aaa-minus-Baa corporate bond yield, is a default-risk measure that affects the cost and availability of external capital and therefore the precautionary motive for holding cash. Model 5 of Panel A shows that for both the 1980s and the 1990s, the value of cash is not associated with credit spread. However, in the 2000s, cash holdings become more valuable when credit spread widens.

As discussed in Hypothesis 4, product market competition intensified during our sample period, and cash holdings are known to have strategic value in the context of product market competition. Our primary measure of competition is the Herfindahl–Hirschman Index (HHI) of industry-level concentration. A lower HHI is associated with more intense competition. In model 6 of Panel A we incorporate this measure to evaluate the impact of industry competition on the value of cash over the three decades of our sample. The results suggest that in the 1980s, competition is not significantly related to the value of cash. However, in the 1990s there is a significant increase in the value of cash for firms operating in more competitive industries. In an untabulated test, we alternatively estimate the effect of competition with the Irvine and Pontiff (2009) measure of industry turnover in a model otherwise identical to model 6. Similar to the HHI finding, we find that higher industry turnover (more intense competition) is associated with a higher marginal value of cash in the 1990s.

While the results in model 6 of Panel A suggest that the effects of product market competition are largely isolated to the decade of the 1990s, we note that this decade experienced a profound increase in competition as measured by both the HHI index and industry turnover. For example, average annual industry turnover virtually quadrupled during the decade, from a low of 0.032 in 1990 to a high of 0.117 in 1999 before dropping precipitously to a low of 0.023 in 2003. In stark contrast, average annual industry competition, as measured by HHI and industry turnover, was essentially unchanged in our sample from 1980 to 1989.

Our sample period witnessed a decrease in diversification in multi-segment firms. As observed in Duchin (2010), multi-segment firms have a lower precautionary demand for cash given imperfect correlations between investment opportunities and cash flows across business segments. As outlined in Hypothesis 5, we extend our analysis to explore whether a decrease in diversification over time also contributes to the secular increase in the marginal value of cash for multi-segment firms.

We follow Duchin (2010) and estimate within-firm diversification as the correlation of cross-segment cash flows and the correlation of cross-segment investment opportunities (Q) for firms reporting multiple segments in Compustat. Given that this methodology requires 10 years of data, our estimates start in 1990. We begin by confirming that the degree of diversification has decreased over time. Specifically, when regressing the annual average correlation in Q on a time trend variable and a constant, the coefficient estimate on the trend variable is 0.114 (t-stat. = 4.42). The average cash-flow correlation also exhibits a significant positive trend using the same specification. One might expect that the decrease in diversification should be less pronounced for firms with fewer number of segments, and we find this is indeed the case. The increase in the average correlation in Q is relatively small for two-segment firms (coeff. = 0.080, t-stat. = 3.56), but nearly twice as large for firms reporting three-or-more segments (coeff. = 0.150; t-stat. = 4.94). Similarly, the increase in cash-flow correlation for two-segment firms is insignificantly different from 0, but is positive and significant for three-or-more-segment firms (coeff. = 0.005; t-stat. = 2.23).

Given the time trend in average correlations, we utilize data on the number of reported business segments to examine whether the reduction in diversification can explain the trend in the marginal value of cash. In model 7 of Panel A in Table 4, we incorporate an indicator variable equal to 1 if a firm reports three or more business segments in a given year, as well as interaction terms between this indicator variable and the change in cash and decade indicators. Multi-segments firms have a lower value of cash in the 1980s, but exhibit a significant increase in the value of cash in the 1990s ($0.31 with a *t*-stat. = 2.13). The change in the value of cash for multi-segment firms is not significant in the 2000s, a result that comports with Duchin (2010) who finds that the reduction in diversification was most pronounced in the 1990s.

As outlined in Hypothesis 6, we also examine whether sentiment is driving investors' valuation of cash over time. In model 8 of Panel A, we find that investor sentiment is weakly positively related to the value of cash in the 1980s (coeff. = 0.050, *t*-stat. = 1.74), becomes slightly more positive in the 1990s (coeff. = 0.103, *t*-stat. = 1.68), but does not change significantly in the 2000s. The coefficient estimates suggest that a 1-standard-deviation increase in sentiment increases the value of cash by $0.07 in the 1980s and $0.22 in the 1990s.

The results in Panel A of Table 4 suggest that, to varying degrees, economic determinants associated with the composition of listed firms, the precautionary demand for cash, and secular trends in credit spreads, product market competition, diversification and investor sentiment all contributed to the secular increase in the value of cash. While the results in Panel A are suggestive, we will be able to draw more conclusive inferences by examining these economic determinants simultaneously. We do so in the next section.

C. The Impact of All Characteristics Simultaneously

In Panel B of Table 4, we evaluate the economic determinants of the value of cash, outlined in Panel A, in a single regression. For ease of exposition we report individual coefficients associated with the determinants by row in the table. We first note that when incorporating all the determinants, the coefficients on $\Delta C_t \times$ DUM_1990 and $\Delta C_t \times$ DUM_2000 become statistically insignificant. This indicates that in sum, the determinants modeled in Panel

B largely explain the secular variation in the value of cash over the sample period. We also note that relative to Panel A of Table 4, the economic inferences in Panel B are largely unchanged with the exception of credit spread and diversification. The effect of credit spread does not increase from the 1980s to the 1990s in Panel A, but does in Panel B. The multi-segment indicator is negatively related to the value of cash in the 1980s in Panel A, but is insignificant in Panel B.

VI. The Speed of Adjustment in Cash Holdings Over Time

Given the significant increase in the demand for precautionary cash holdings, a coincident increase in capital market frictions might explain why the value of cash also increased over time. Despite improvements in financial technologies, Bates et al. (2009) observe that U.S. industrial firms typically hold less cash than predicted during the sample period. Given this observation, and our findings that the value of cash has risen consistently over time, it is likely that the demand for corporate liquidity is, on average, outstripping the average firm's ability to generate surpluses from internal cash flows, external capital raising, and asset sales. We estimate a partial adjustment model to examine the SOA in firm cash holdings. Partial adjustment models are commonly applied in the capital structure literature and have been used to consider the dynamics of corporate liquidity.[19] For example, Dittmar and Duchin (2010) employ a partial adjustment model to study the SOA of cash holdings over the corporate life cycle, and Gao, Harford, and Li (2013) compare the SOA of cash holdings for public and private firms.

We estimate the partial adjustment model of cash as:

$$(2)\quad \mathrm{CASH}_{i,t} - \mathrm{CASH}_{i,t-1} = \alpha \times (\mathrm{CASH}^{*}_{i,t} - \mathrm{CASH}_{i,t-1}) + \epsilon_{i,t}\ ,$$

[19] In the capital structure literature see Taggart (1977), Auerbach (1985), Shyam-Sunder and Myers (1999), Fama and French (2002), Welch (2004), Flannery and Rangan (2006), Lemmon, Roberts, and Zender (2008), and Chang and Dasgupta (2009), among others.

where $\text{CASH}^*_{i,t}$ denotes the target cash ratio at time t, and α is the target adjustment coefficient. The coefficient α is bounded between 0 (no adjustment) and 1 (perfect adjustment). Rearranging the terms of equation (2) gives us:

$$(3) \quad \text{CASH}_{i,t} = (1-\alpha) \times \text{CASH}_{i,t-1} + \alpha \times \text{CASH}^*_{i,t} + \epsilon_{i,t} \, .$$

We can implement equation (3) through the following regression model:

$$(4) \quad \text{CASH}_{i,t} = \gamma_0 + \gamma_1 \times \text{CASH}_{i,t-1} + \beta \times x_{i,t} + \epsilon_{i,t} \, ,$$

where $x_{i,t}$ is a set of firm characteristics that predict $\text{CASH}^*_{i,t}$, the target level of cash, estimated as a function of cash-flow volatility, the market-to-book ratio, firm size, cash flow, net working capital, capital expenditure, financial leverage, R&D, a dividend dummy, and acquisition expenses.

In equation (4), the SOA of cash is $(1 - \gamma_1)$. Adjustment costs may be asymmetric for firms with positive or negative excess cash. Firms with positive excess cash may adjust the cash level downward through capital spending, acquisitions, dividend payments, and share repurchases. Adjustments for firms with negative excess cash are a function of the cost of external financing. To estimate the difference, we include a measure of excess cash (EX_CASH) into the SOA model as follows:

$$(5) \quad \text{CASH}_{i,t} = \gamma_0 + \gamma_1 \times \text{CASH}_{i,t-1} + \gamma_2 \times \text{EX\_CASH}_{i,t-1} \times \text{CASH}_{i,t-1} +$$

$$\gamma_3 \times \text{EX\_CASH}_{i,t-1} + \beta \times x_{i,t} + \epsilon_{i,t} \, ,$$

where the SOA is $[1 - (\gamma_1 + \gamma_2 \times \text{EX\_CASH}_{i,t-1})]$. We expect γ_2 to be negative, that is, as excess cash increases, the SOA becomes larger (faster). We calculate excess cash as described in Section III.

In Panel A of Table 5 we summarize the results of a partial adjustment model using an OLS specification of equation (5).[20] We first estimate the model for the full sample period using a continuous measure of excess cash (model 1) and quartiles of excess cash (model 2). Excess-cash quartiles are constructed by first sorting firms into two subgroups based on whether a firm has positive or negative excess cash. For each of the two subgroups, we again sort firms into two halves based on their excess-cash ratio. By construction the top quartile group includes firms with the greatest cash surplus, while those in the bottom quartile have the greatest cash deficit.

[Insert Table 5 about here]

In both models 1 and 2 in Panel A of Table 5, the coefficient estimate on the interaction term between the lagged level of cash and excess cash is negative. This implies that the SOA is faster for higher levels of excess cash. Recall that the SOA in equation (5) can be computed as $[1 - (\gamma_1 + \gamma_2 \times \mathrm{EX\_CASH}_{i,t-1})]$. Given the coefficients from model 2, the SOA for firms in the top quartile of excess cash is 0.339 = [1 – (0.689 – 0.007 × 4)]. In contrast, firms in the bottom quartile of excess cash exhibit an SOA of 0.318.[21] These results support the notion that firms face greater frictions when attempting to close a cash deficit than when they seek to disgorge surplus cash.

While the SOA results in models 1 and 2 of Table 5 are suggestive, we also consider whether the SOA has changed over time for firms with different levels of excess cash. In models 3 and 4 we interact decade indicators with lagged cash and our continuous and quartile sorts on excess cash. In both models the coefficient estimates on LAGGED CASH × DUM_1990 and

[20] In studies of target capital structure, firm fixed-effects models have been implemented to control for the effect of unobservable firm heterogeneity. A firm fixed-effects specification of equation (5) creates a dynamic panel bias, which Blundell and Bond (1998) show can be corrected using generalized method of moments (GMM). In untabulated specifications we also estimate a GMM model and obtain similar results as those of the OLS specification. In GMM specifications, we first use lagged 2 years and older *x* variables as instruments for the lagged cash level and obtain similar results as those from the OLS specification. We also utilize lagged 3 years and older, lagged 4 years and older, and lagged 5 years and older *x* variables as instruments and again obtain equivalently significant results.

[21] A GMM specification yields qualitatively similar results. For firms in the highest excess cash quartile, the SOA is 0.415; for firms in the lowest quartile, the SOA is 0.361.

LAGGED CASH × DUM_2000 are significantly positive, indicating that the SOA has decreased over the last two decades. The coefficient estimates on the three-way interactions between lagged cash, excess cash, and decade indicators are significantly negative, suggesting that the reduction in the SOA has been more pronounced for firms with negative excess cash.

To facilitate the economic interpretation of the results in Panel A of Table 5, Panel B summarizes the SOA for each quartile of excess cash over time based on the results in model 4 of Panel A. The bottom row of the table conveys the difference between the SOA in the 1980s and the 2000s for each quartile. While the SOA has decreased over time for all excess-cash quartiles, the differences are more pronounced for firms with negative excess cash. For example, the SOA for the bottom excess-cash quartile declined by 39% from the 1980s to the 2000s (from 0.467 to 0.287), compared to a decline of 21% (from 0.377 to 0.299) for the top quartile. The rightmost column shows that in the 1980s, the top excess-cash quartile has a slower SOA than the bottom quartile. This difference, however, reverses in the 1990s and 2000s, suggesting that the impact of financial constraints for firms with negative excess cash were more pronounced in these later decades. These findings cohere with the earlier observation in Figure 2 that the value of cash for negative-excess-cash firms has increased much more significantly than the value for positive-excess-cash firms since the mid-1990s.

In Table 6 we explicitly examine the effect of financial constraints on the speed of adjustment over time in specifications equivalent to model 4 of Panel A in Table 5. We use the five alternative proxies for financial constraints described in Section V.B. Constrained firms are defined by above-median SA and Whited–Wu indices as well as firms without a bond or commercial credit rating, and firms that do not pay a regular dividend. For each definition of financial constraint, we run the SOA regression for the unconstrained subsample in the odd-numbered columns, and for the constrained subsample in the even-numbered columns of Table 6.

[Insert Table 6 about here]

In columns 1 and 2 of Table 6 we estimate the speed of adjustment for firms with low and high financial constraints, respectively, as defined by the SA index. When financial constraints are low, the interaction terms between lagged cash, excess cash, and decade indicators are insignificant, suggesting that for unconstrained firms the SOA is not related to excess cash levels and this relation is unchanged over time. In column 2, however, the interaction terms between lagged cash, excess cash, and decade indicators are statistically significant, and negative signs on the coefficients for the three-way interaction terms indicate that the SOA has slowed down significantly for constrained firms with cash deficits in the 1990s and 2000s. We observe similar results across the four alternative proxies for financial constraints, suggesting that the deceleration in the SOA for firms with cash deficits is largely driven by firms with greater financial constraints. For firms with lower financial constraints, there is little evidence that the SOA has changed over time or is related to the level of excess cash.[22]

The application of partial adjustment models has been questioned for its ability to truly detect active adjustment in capital structure. For example, Chang and Dasgupta (2009) show that target models in capital structure suffer from mechanical mean reversion in debt level that may obtain even in the absence of a target capital structure. This critique suggests that we exercise some caution in interpreting our results concerning the speed of adjustment in cash holdings. This said, we are not aware of a comparable mechanical channel for mean reversion in the cash ratio. Furthermore, the emphasis of our analysis is less on estimating the precise SOA than it is on evaluating the change in the SOA over time and between excess cash subsamples. Thus, even with some degree of mean reversion that is stable over time and across firms, our evidence should capture the differences in active adjustment over time.

Caveats notwithstanding, the findings in this section suggest that capital market frictions for firms with cash deficits have increased substantially over time. While the demand for

[22] It is possible that firms with negative excess cash are primarily those facing greater financial constraints. However, we do not find this to be the case. The correlations between excess cash and our five measures of financial constraints are all below 6%.

liquidity increased steadily over our sample period, the average firm's ability to access sources of liquidity and maintain cash reserves failed to keep pace. Our results suggest that this problem is particularly acute for firms with larger cash deficits and greater financial constraints, providing a reasonable economic explanation for the increase in marginal value of cash during the sample period.

VII. Agency Conflicts and the Secular Increase in the Value of Cash

In this section, we consider whether the time trend in the marginal value of cash can be attributed to an improvement in corporate governance over time. The literature has established a strong relation between the quality of corporate governance and the marginal value of cash for the cross section of firms. For example, Dittmar and Mahrt-Smith (2007) study a sample of U.S. firms and find that the value of cash is lower for firms with less institutional ownership and fewer shareholder rights as measured by the Gompers et al. (2003) G-index. Similarly, Pinkowitz et al. (2006) find that the value of cash is lower in countries with poor investor protection.

To date, there is little evidence that corporate governance can account for secular changes in the value of cash. To this issue, Bates et al. (2009) conclude that the increase in the level of cash over the sample period is not a function of agency problems in firms. We first consider the influence of large institutional shareholders on the marginal value of cash over time. We follow Cremers and Nair (2005) and measure the cumulative percentage of a firm's shares held in 5%+ blocks by institutional investors using the Thompson Reuters Institutional Holdings (13F) database (5%_BLOCK). We estimate a specification that is identical to Panel B of Table 4 and add our block holdings measure and its interaction terms with the change in cash and the decade indicators. The results of the regression are summarized in Table 7. For brevity we do not tabulate the coefficients associated with the other variables and the intercept. The coefficient estimates indicate that in the 1980s and the 2000s, institutional block holdings have a negligible impact on the marginal value of cash. For the 1990s, the effect of block holdings is positive (coeff. = 0.957, t-stat. = 1.75), but the economic effect is relatively small where a one standard deviation increase in block holdings is associated with an \$0.11 increase in the marginal value of cash.

[Insert Table 7 about here]

We next consider time variation in the marginal value of cash associated with the G-index. We follow Gompers et al. (2003) and compute the G-index as a cumulative count of 24 different governance provisions for firms covered by RiskMetrics during the years 1990-2007. We then sort these firms annually into terciles of the G-index. Employing equation (1), we estimate the value of cash annually for each tercile and plot the estimates in Figure 12. The results are consistent with Dittmar and Mahrt-Smith (2007) where the marginal value of cash is higher for firms in the lower G-index terciles. The regression point estimates suggest that between 1990 and 2007, the value of cash for firms in the lowest tercile is, on average, \$0.353 higher than the value of cash for firms in the highest tercile. The difference in the marginal value of cash between the low and high terciles, however, does not trend upward over time, suggesting that the G-index does little to explain the secular increase in the value of cash.

[Insert Figure 12 about here]

VIII. Conclusion

We document a secular increase in the value of cash by U.S. non-financial firms from the 1980s to the 2000s. For the average firm in the sample, an additional dollar of cash holdings is valued at \$0.61 in the 1980s, \$1.04 in the 1990s, and \$1.12 in the 2000s. We find that the increase in the value of cash obtains across a variety of firm characteristics including firm size, the investment opportunity set, financial constraints, and the volatility of cash flow. The secular increase in the value of cash is robust to alternative valuation models, the inclusion of conventional control variables in the literature, and various alternative explanations.

The increase in the value of cash can be partially attributed to IPO firms and changes in the composition of the market over time, but is also a function of changing firm fundamentals that characterize the precautionary demand for cash. In the 1990s, the increase in the value of cash is predominantly driven by a firm's investment opportunity set, the volatility of its cash flow, and product market competition. For the 2000s, the increase in the value of cash is largely explained by credit market risk. Finally, the decline in corporate diversification and higher investor sentiment help explain the increase in the marginal value of cash from the 1980s through the 1990s.

The secular increase in the value and level of cash, coupled with a coincident increase in the demand for precautionary reserves, is consistent with the presence of significant adjustment costs for firms with liquidity deficits. To identify the extent of financing frictions over time, we estimate a partial adjustment model for cash holdings during our sample period. Our results show that the speed of adjustment in cash holdings has declined significantly over time, and most acutely for firms with cash deficits. Furthermore, the decline in adjustment speed is largely driven by firms facing greater financial constraints. This evidence provides an important economic link between the increasing demand for cash balances over time and the trend in the value of these balances documented in this paper. Understanding the determinants of this time series trend in financing frictions remains an important question for future research in corporate liquidity management.

Finally, our research suggests that corporate governance features cannot explain much of the increase in the marginal value of cash over the sample period. Consistent with prior research, we find that the marginal value of cash is higher when corporate ownership includes large institutional blockholders, and when a firm maintains relatively strong shareholder rights. The presence of institutional investors, however, explains only a small fraction of the increase in the value of cash in the 1990s, while an index of shareholder rights is uncorrelated with the time trend in the value of cash.

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Figure 1: Value of Cash Holdings: Annual Estimates and 5-Year and 10-Year Moving Averages

Figure 1 shows the annual estimates and the 5-year and 10-year moving averages of the value of an additional dollar of cash holdings from 1980 to 2009. The sample covers all Compustat firm-year observations that have the requisite variables to estimate the value of cash. The sample period is from 1980 to 2009. Financial firms (SIC codes between 6000 and 6999) and utility firms (SIC codes between 4900 and 4999) are excluded from the sample. The final sample generates a panel of 12,845 unique firms with 108,357 firm-year observations. We run the regression each year based on the first model specification below following Faulkender and Wang (FW) (2006). All variables are defined in the text and Table 1. γ_1 is the estimate of the average value of \$1 marginal cash holding. We obtain γ_1 each year and plot its annual value and 5- and 10-year moving averages in the figure. In addition, we obtain β_1 as an alternative estimate of the average value of \$1 marginal cash holding each year based on the second model specification following Pinkowitz et al. (PSW) (2006) and plot the 10-year moving average in the figure. V is firm market value as defined in PSW. All variables in the PSW regression are scaled by firm book value of assets.

Faulkender and Wang (2006):

$$r_{i,t} - R_{i,t}^{B} = \gamma_0 + \gamma_1 \frac{\Delta C_{i,t}}{M_{i,t-1}} + \gamma_2 \frac{\Delta E_{i,t}}{M_{i,t-1}} + \gamma_3 \frac{\Delta \mathrm{NA}_{i,t}}{M_{i,t-1}} + \gamma_4 \frac{\Delta \mathrm{RD}_{i,t}}{M_{i,t-1}} + \gamma_5 \frac{\Delta I_{i,t}}{M_{i,t-1}} + \gamma_6 \frac{\Delta D_{i,t}}{M_{i,t-1}} + \gamma_7 \frac{C_{i,t-1}}{M_{i,t-1}} + \gamma_8 L_{i,t} + \gamma_9 \frac{\mathrm{NF}_{i,t}}{M_{i,t-1}} + \varepsilon_{i,t}\,.$$

Pinkowitz et al. (2006):

$$V_{i,t} = \beta_0 + \beta_1 dC_{i,t} + \beta_2 dC_{i,t+1} + \beta_3 E_{i,t} + \beta_4 dE_{i,t} + \beta_5 dE_{i,t+1} + \beta_6 d\mathrm{NA}_{i,t} + \beta_7 d\mathrm{NA}_{i,t+1} + \beta_8 \mathrm{RD}_{i,t} + \beta_9 d\mathrm{RD}_{i,t} + \beta_{10} d\mathrm{RD}_{i,t+1} + \beta_{11} I_{i,t} + \beta_{12} dI_{i,t} + \beta_{13} dI_{i,t+1} + \beta_{14} D_{i,t} + \beta_{15} dD_{i,t} + \beta_{16} dD_{i,t+1} + \beta_{17} dV_{i,t+1} + \varepsilon_{i,t}\,.$$

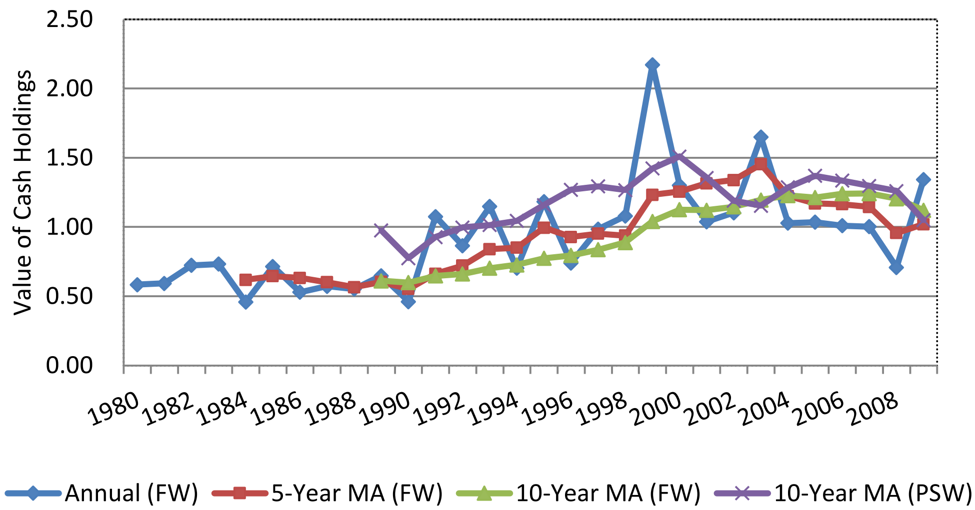

Figure 2: Value of Cash by Top and Bottom Excess-Cash Terciles

Figure 2 presents the 10-year moving average of the value of cash for top and bottom excess-cash terciles. We run Fama–MacBeth (1973) regressions of cash-to-total-assets ratio on the determinants of cash level over the 30-year sample period, and use the average coefficients to predict the level of cash. Excess cash is the difference between actual cash holdings and predicted cash holdings. Every year, we sort firms into terciles based on its excess-cash level at fiscal year *t* (the middle terciles is discarded), and estimate the value of cash for each excess-cash tercile. We then obtain the 10-year moving average of the value of cash for each excess-cash tercile.

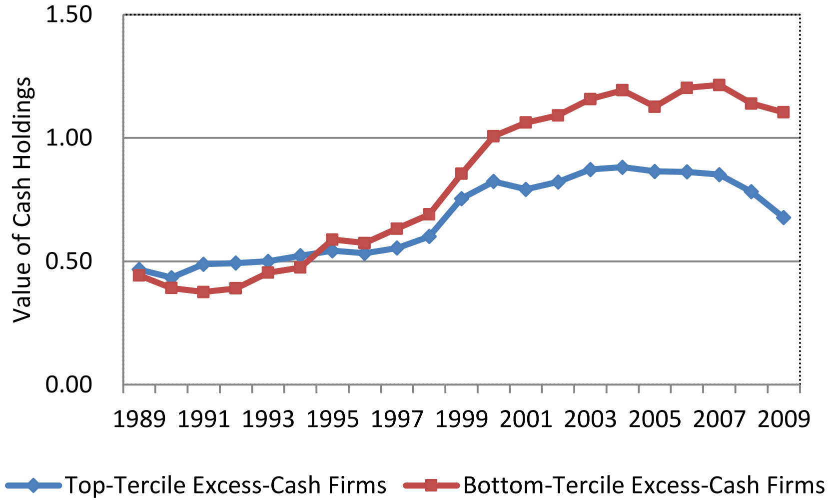

Figure 3: Value of Cash by Size Quintiles

Figure 3 presents the 10-year moving average of the value of cash for size quintiles. We define size as the book value of total assets (Compustat item AT). Every year, we sort firms into quintiles based on its size at the beginning of fiscal year t, and estimate the value of cash for each quintile. We then obtain the 10-year moving average of the value of cash for each size quintile.

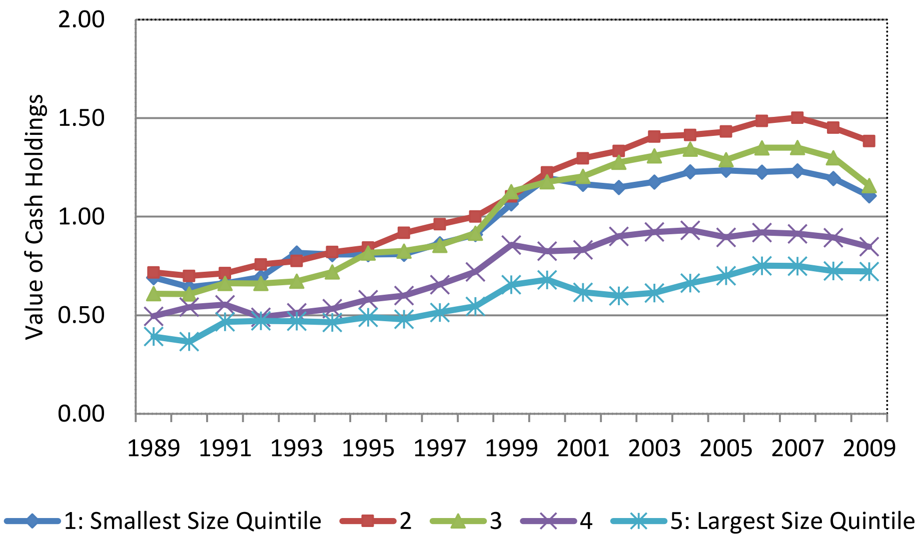

Figure 4: Value of Cash by the Level of R&D Activities

Figure 4 presents the 10-year moving average of the value of cash based on the level of R&D activities. We measure R&D by Compustat item XRD. We normalize R&D by the firm's book value of assets. Every year, we sort firms into zero, low, and high R&D activities based on the level of R&D at the beginning of the fiscal year. Specifically, for non-zero R&D firms, we use the median R&D ratio as the cutoff point and we separate the firms into low- or high-R&D groups. We estimate the value of cash for each R&D group and then compute the 10-year moving average for each R&D group. The sample covers all Compustat firm-year observations that have the requisite variables to estimate the value of cash. The sample period is from 1980 to 2009. Financial firms (SIC codes between 6000 and 6999) and utility firms (SIC codes between 4900 and 4999) are excluded from the sample.

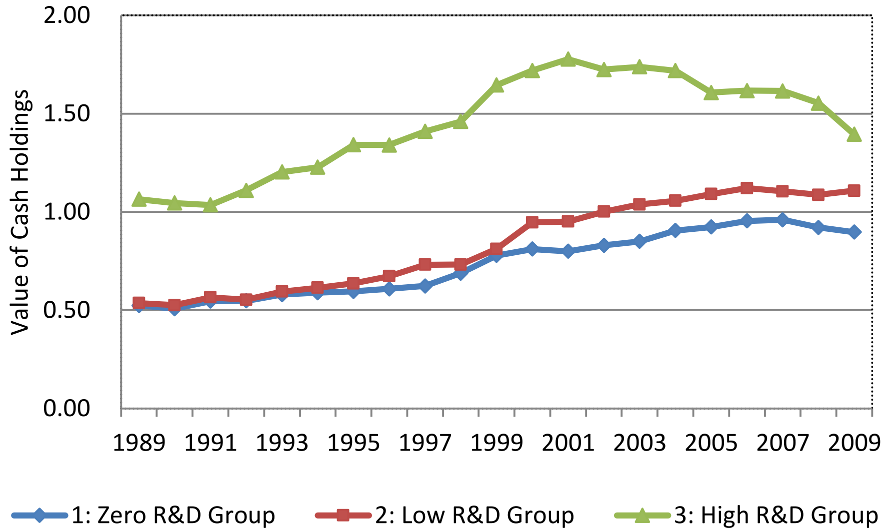

Figure 5: Value of Cash by Technology and Non-technology Firms

Figure 5 presents the 10-year moving average of the value of cash for technology and non-technology firms. We classify technology firms following the method in Loughran and Ritter (2004). Every year, we sort firms into technology and non-technology groups based on whether a firm is a technology firm at the beginning of the fiscal year and estimate the corresponding value of cash. We then obtain the 10-year moving average of the value of cash for each group,

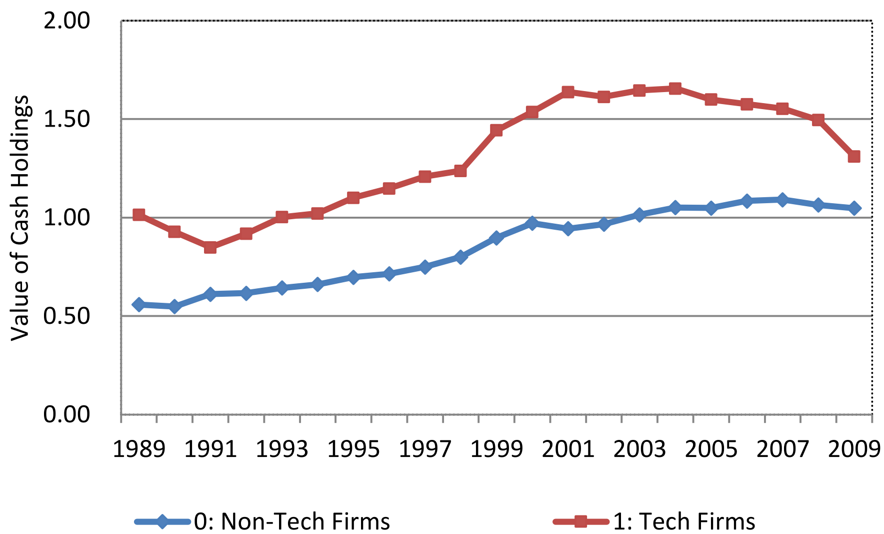

Figure 6: Value of Cash by MB Ratio Quintiles

Figure 6 presents the 10-year moving average of the value of cash for MB quintiles. We define MB ratio as the book value of assets minus the book value of equity plus the market value of equity over the book value of assets (Compustat items ((AT – CEQ + (PRCC_F × CSHO)) / AT). Every year, we sort firms into quintiles based on MB at the beginning of the fiscal year, and compute the corresponding value of cash. We then obtain the 10-year moving average of the value of cash for each MB quintile.

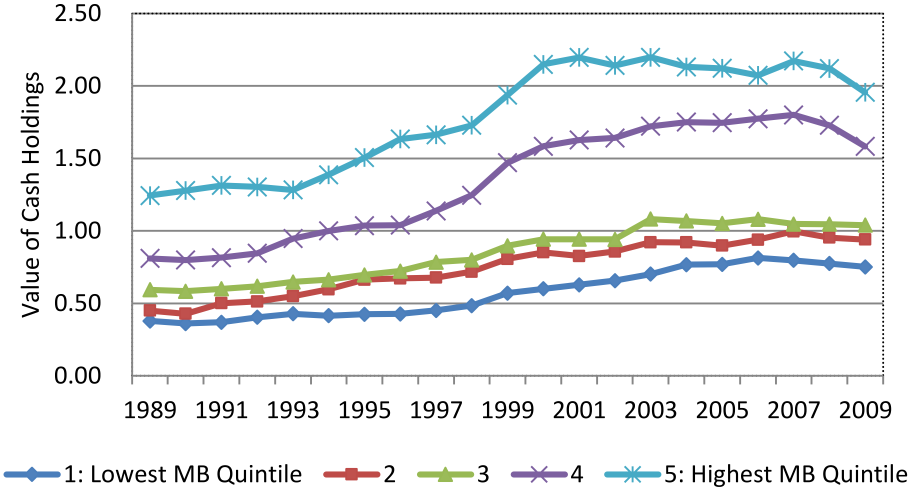

Figure 7: Value of Cash by Cash-flow Volatility Quintiles

Figure 7 presents the 10-year moving average of the value of cash for quintiles sorted on cash-flow volatility. We define cash flow as cash flow over net assets (Compustat items (OIBDP – XINT – TXT – DVC) / (AT – CHE)). For each firm in a given year, we compute the standard deviation of cash flow for the past twenty years, requiring a minimum of 5 years' observations. We then average the cash-flow standard deviation for each 2-digit SIC industry, and use the industry cash-flow standard deviation as the measure of cash-flow volatility. We sort firms each year into cash-flow volatility quintiles. For each quintile, we estimate the value of cash and compute the 10-year moving average.

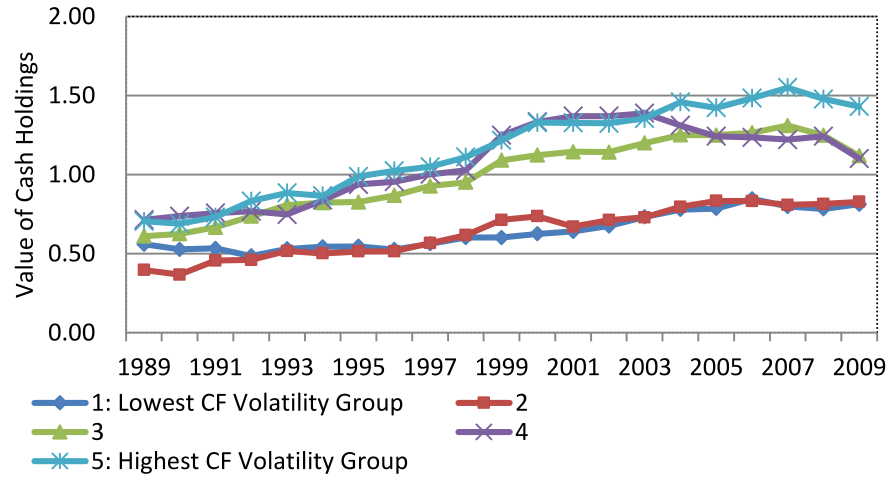

Figure 8: Value of Cash by SA Financial Constraint Index

Figure 8 presents the 10-year moving average of the value of cash based on SA financial constraint index. Following Hadlock and Pierce (2010), we calculate the SA index as (– 0.737 × Size) + (0.043) × Size^2 – (0.040 × Age). Higher SA index value indicates greater financial constraints. Every year, we sort firms into quintiles based on the SA index value at the beginning of the fiscal year, and compute the corresponding value of cash. We then obtain the 10-year moving average of the value of cash for each SA index quintile.

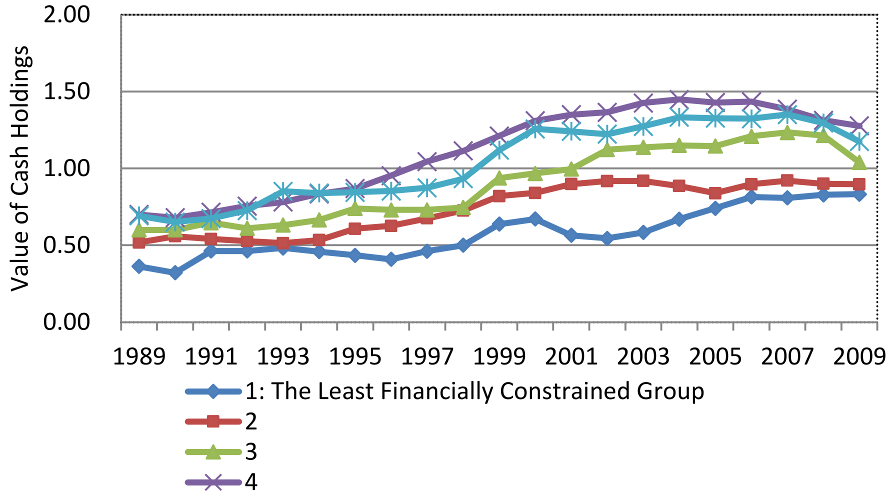

Figure 9: Value of Cash by Credit Rating

Figure 9 presents the 10-year moving average of the value of cash for two groups: those with investment grade credit rating and those with junk or no crediting rating (Almeida et al. (2004). Every year, we sort firms into the two groups based on credit rating status at the beginning of the fiscal year and estimate the corresponding value of cash. We then obtain the 10-year moving average of the value of cash for each group.

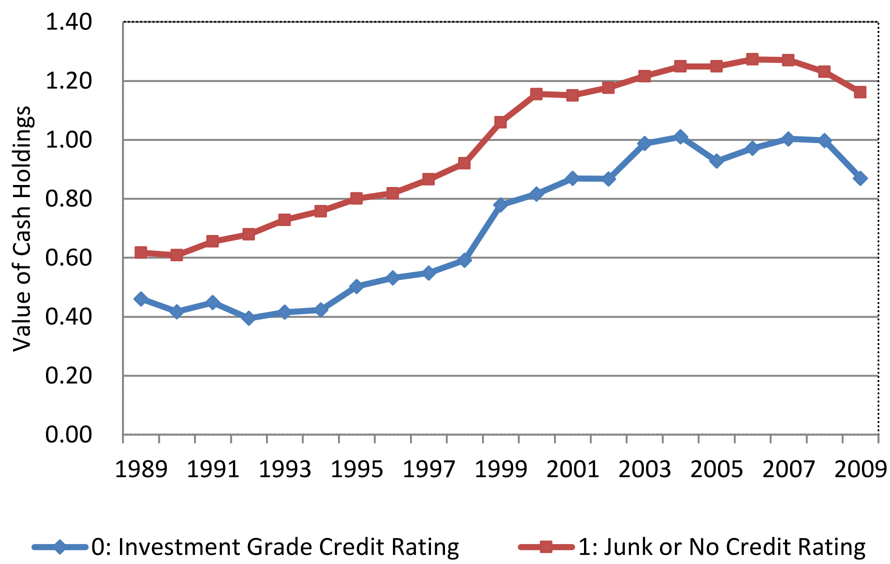

Figure 10: Value of Cash by Foreign Income

Figure 10 presents the 10-year moving average of the value of cash based on the level of foreign income activities. We measure foreign income by Compustat item PIFO and normalize foreign income by firm book value of assets. Every year, we sort firms into zero, low, and high foreign income activities based on the level of foreign income at the beginning of the fiscal year. Specifically, for non-zero foreign income firms, we use the median foreign income ratio as the cutoff point and separate the firms into low- or high-foreign income groups. We estimate the value of cash for each foreign income group and then compute the 10-year moving average for each foreign income group.

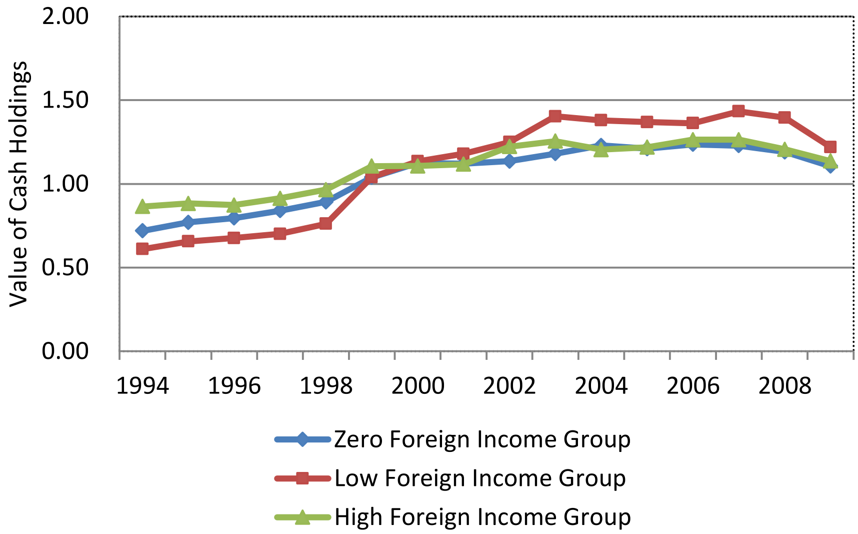

Figure 11: Value of Cash by IPO Cohorts.

Figure 11 presents the 10-year moving average of the value of cash for cohorts of firms based on IPO dates. We obtain IPO years from Thomson's SDC database. When we cannot obtain IPO years from SDC, we assume the IPO year is the first year the firm appears in Compustat minus 1. We adopt this minus-one-year adjustment because we observe that for the majority of firms that we can identify the IPO years from SDC, the IPO year is the year prior to the firm's first year in Compustat. For a given year, we then assign a firm as an IPO firm if it has gone public within the prior five years. The 1960s cohort contains firms that have gone public prior to 1970. The 1970s cohort covers firms that become public from 1970 to 1974. The 1975s cohort covers firms that become public from 1975 to 1979, and so on. For each cohort, we estimate the value of cash every year and plot the 10-year moving average.

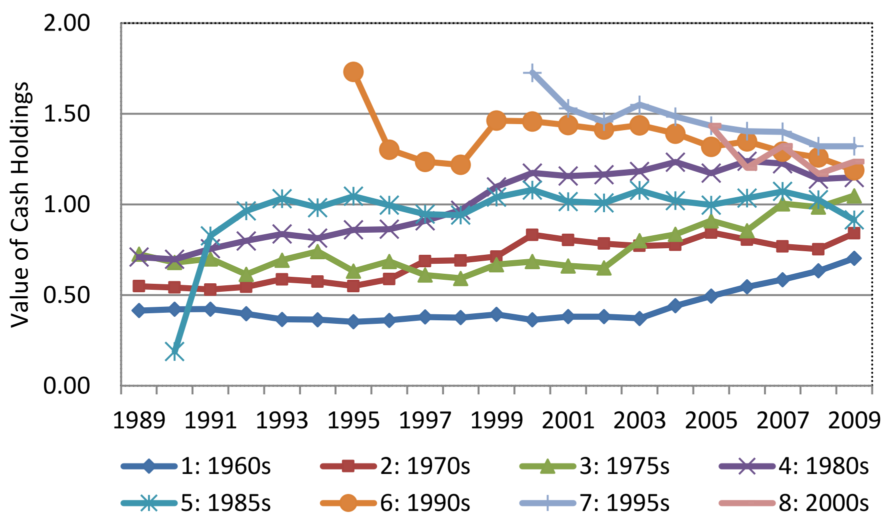

Figure 12: Value of Cash by G-Index

Figure 12 plots the annual estimate of the value of cash for terciles sorted on G-index. Every year, we rank firms into three groups based on their corresponding G-index. Also, following Gompers et al. (2003), we use the most recent G-index for the years when the report is not available. For each tercile and year, we estimate the value of cash using the baseline equation. The sample covers all Compustat firm-year observations that have the requisite variables to estimate the value of cash. The sample period is from 1980 to 2007.

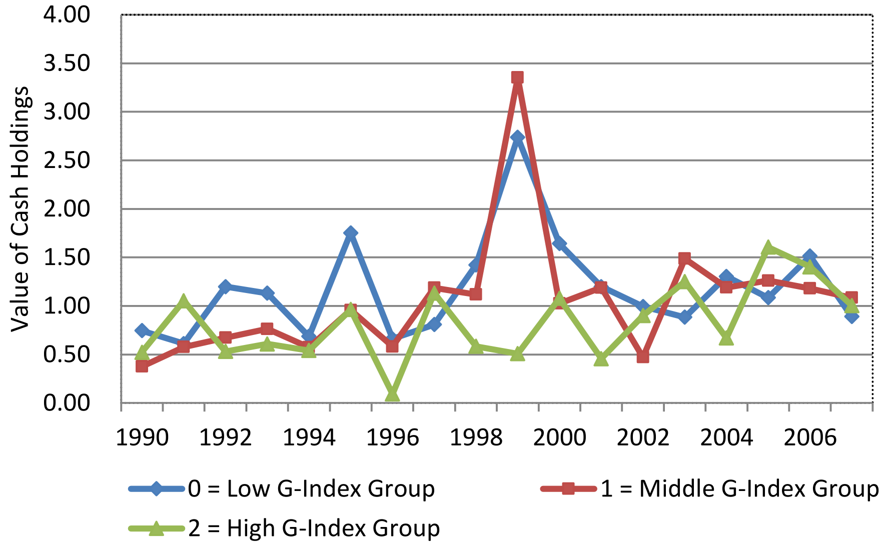

Table 1: Summary Statistics

Table 1 presents summary statistics for all variables in our sample. We include U.S. publicly traded companies from 1980 to 2009 except financial firms (SIC codes between 6000 and 6999) and utility firms (SIC codes between 4900 and 4999). $\mathrm{r_{i,t}} - \mathrm{R^{B}_{i,t}}$ is the excess return in which $r_{i,t}$ is a stock's return in a given fiscal year *t* and $R^{B}_{i,t}$ is the stock's benchmark portfolio return for the same corresponding period. AT is book value of total assets. MB is market-to-book ratio calculated by the book value of assets minus the book value of equity plus the market value of equity over the book value of assets. RD is research and development expenses over book value of total assets. *L* is market leverage computed by long-term debt plus short–term debt over the sum of long-term debt, short–term debt, and market value of equity. FOREIGN_INCOME is foreign income over book value of total assets. DIVIDEND_DUMMY equals to 1 for dividend-paying firms. SA_INDEX is constructed following Hadlock and Pierce (2010). CASH-FLOW_VOLATILITY is the standard deviation of the cash flow-to-total assets ratio of 2-digit SIC industries using past 20 years observations. CREDIT_RATING_DUMMY is constructed following Almeida et al. (2004). Firms with junk or no credit rating are financially constrained and coded as 1. Firms with an investment grade credit rating are coded as 0. TECH_DUMMY is defined as in Loughran and Ritter (2004). Tech firms are coded as 1. CREDIT_SPREAD is defined as Aaa-minus-Baa corporate bond yield. HHI measures industry concentration of sales. MULTISEG_DUMMY is equal to 1 if a firm has three or more business segments. SENTIMENT measures investor sentiment as detailed in the text. 5%_BLOCK is the total block institutional ownership (i.e., the sum of > 5% institutional holdings), from Thomson Reuters Institutional Holdings (13F) Database. *C* is cash plus marketable securities. NF is net financing which is total equity issuance minus repurchase plus debt issuance minus redemption. *D* is common dividends paid. *E* is earnings before extraordinary items plus interest, deferred tax credits, and investment tax credits. *I* is interest expense. NA is total assets minus cash. Δ indicates the change from the previous year. All variables are winsorized at the top and bottom one percentile. *C*, NF, and all the change variables (denoted by Δ) are scaled by lagged market value of equity. The final sample generates a total of 12,845 unique firms with 108,357 firm-year observations.

Variables	Mean	Std. Dev.	Q1	Q2	Q3
$r_{i,t} - R^B_{i,t}$	0.058	0.694	-0.330	-0.046	0.268
AT	2061	12439	30	124	624
MB	1.963	2.342	1.031	1.366	2.068
RD	0.047	0.123	0.000	0.000	0.048
L	0.234	0.231	0.028	0.168	0.376
FOREIGN_INCOME	0.005	0.040	0.000	0.000	0.000
DIVIDEND_DUMMY	0.373	0.484	0.000	0.000	1.000
SA_INDEX	-2.842	0.718	-3.340	-2.928	-2.399
CASH-FLOW_VOLATILITY	0.111	0.087	0.050	0.081	0.140
CREDIT_RATING_DUMMY	0.793	0.405	1.000	1.000	1.000
TECH_DUMMY	0.215	0.411	0.000	0.000	0.000
CREDIT_SPREAD	1.081	0.437	0.742	0.920	1.312
HHI	0.071	0.069	0.035	0.051	0.075
MULTISEG_DUMMY	0.297	0.457	0.000	0.000	1.000
SENTIMENT	0.020	1.408	-0.753	0.070	0.739
5%_BLOCK	0.108	0.125	0.000	0.068	0.174
C	0.176	0.249	0.033	0.092	0.216
NF	0.042	0.255	-0.031	0.001	0.069
ΔC	0.013	0.157	-0.031	0.002	0.043
ΔD	0.000	0.009	0.000	0.000	0.000
ΔE	0.023	0.287	-0.036	0.008	0.049
ΔI	0.002	0.033	-0.002	0.000	0.005
ΔNA	0.067	0.512	-0.043	0.043	0.174
ΔRD	0.000	0.026	0.000	0.000	0.002

Table 2: Value of Cash: Annual Estimates and 5-year and 10-year Moving Averages

Table 2 presents the annual estimates of value of cash (γ_1), along with the 5-year and 10-year moving averages. We use the following regression to estimate the value of cash each year:

$$r_{i,t} - R^B_{i,t} = \gamma_0 + \gamma_1 \frac{\Delta C_{i,t}}{M_{i,t-1}} + \gamma_2 \frac{\Delta E_{i,t}}{M_{i,t-1}} + \gamma_3 \frac{\Delta \mathrm{NA}_{i,t}}{M_{i,t-1}} + \gamma_4 \frac{\Delta \mathrm{RD}_{i,t}}{M_{i,t-1}} + \gamma_5 \frac{\Delta I_{i,t}}{M_{i,t-1}} + \gamma_6 \frac{\Delta D_{i,t}}{M_{i,t-1}} + \gamma_7 \frac{C_{i,t-1}}{M_{i,t-1}} + \gamma_8 L_{i,t} + \gamma_9 \frac{\mathrm{NF}_{i,t}}{M_{i,t-1}} + \varepsilon_{i,t} ,$$

where $r_{i,t} - R^B_{i,t}$ is the excess return in which $r_{i,t}$ is a stock's return in a given fiscal year and $R^B_{i,t}$ is the stock's benchmark portfolio return for the corresponding period. The independent variables are changes in firm characteristics including Cash (*C*), Earnings (*E*), Net assets (NA), R&D (RD), Interest expense (*I*), Dividends (*D*), Leverage (*L*), and Net financing (NF). All variables except leverage and excess return have been deflated by lagged market value of equity of the firm. The subscripts *t*-1 and *t* indicate whether the value of the variable is at the beginning or the end of fiscal year *t*. All variables are winsorized at the top and bottom one percentile to mitigate the impact of extreme outliers. The sample period is from 1980 to 2009. Sample size is 108,357 firm-year observations.

Year	Annual Value-of-Cash Estimate (γ_1)	5-Year Moving Average	10-Year Moving Average
1980	0.584		
1981	0.593		
1982	0.724		
1983	0.733		
1984	0.458	0.618	
1985	0.715	0.645	
1986	0.529	0.632	
1987	0.572	0.601	
1988	0.552	0.565	
1989	0.648	0.603	0.611
1990	0.462	0.553	0.599
1991	1.076	0.662	0.647
1992	0.865	0.721	0.661
1993	1.148	0.840	0.702
1994	0.702	0.851	0.727
1995	1.183	0.995	0.774
1996	0.741	0.928	0.795
1997	0.985	0.952	0.836
1998	1.079	0.938	0.889
1999	2.173	1.232	1.041
2000	1.301	1.256	1.125
2001	1.038	1.315	1.121
2002	1.104	1.339	1.145
2003	1.649	1.453	1.195
2004	1.028	1.224	1.228
2005	1.035	1.171	1.213
2006	1.010	1.165	1.240
2007	1.003	1.145	1.242
2008	0.709	0.957	1.205
2009	1.342	1.020	1.122

Table 3: Baseline Regressions: The Increase in the Value of Cash Over Time

Table 3 shows the regression result of excess return on changes in firm characteristics. All variables except leverage and excess return have been deflated by lagged market value of equity of the firm. Model 1, our baseline regression, includes Cash (*C*), Earnings (*E*), Net assets (NA), R&D (RD), Interest expense (*I*), Dividends (*D*), Leverage (*L*), and Net financing (NF). Model 2 adds an interaction term between change in cash and 1990s and 2000s decade indicators. model 3 further includes interaction terms between change in cash and lagged level of cash and leverage, and 3-way interactions with decade indicators. The subscripts *t*-1 and *t* indicate whether the value of the variable is at the beginning or the end of fiscal year *t*. All variables are winsorized at the top and bottom one percentile to mitigate the impact of extreme outliers. The sample period is from 1980 to 2009. Sample size is 108,357 firm-year observations. In parentheses are *t*-statistics based on standard errors that are robust to heteroskedasticity and firm and year clustering. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Variables	Base 1	Time Dummy 2	Cash Level and Leverage 3
ΔC_t	1.078***	0.711***	1.092***
	(10.66)	(22.40)	(12.33)
ΔC_t × DUM_1990		0.498***	0.931***
		(2.95)	(2.64)
ΔC_t × DUM_2000		0.552***	1.045***
		(5.10)	(2.74)
$\Delta C_t * C_{t-1}$			-0.262
			(-1.46)
$\Delta C_t * C_{t-1}$ × DUM_1990			-0.581**
			(-2.10)
$\Delta C_t * C_{t-1}$ × DUM_2000			-0.636
			(-1.58)
$\Delta C_t * L_t$			-0.962***
			(-8.37)
$\Delta C_t * L_t$ × DUM_1990			-1.382**
			(-2.23)
$\Delta C_t * L_t$ × DUM_2000			-0.905***
			(-2.71)
ΔE_t	0.478***	0.477***	0.471***
	(11.99)	(11.98)	(12.26)
ΔNA_t	0.216***	0.216***	0.226***
	(21.53)	(21.16)	(24.13)
ΔRD_t	0.272	0.218	0.207
	(0.47)	(0.37)	(0.39)
ΔI_t	-1.854***	-1.859***	-1.759***
	(-6.56)	(-6.53)	(-6.27)
ΔD_t	2.601***	2.662***	2.505***
	(4.10)	(4.20)	(3.68)
C_{t-1}	0.460***	0.467***	0.387***
	(4.97)	(5.01)	(4.53)
L_t	-0.522***	-0.522***	-0.515***
	(-8.71)	(-8.83)	(-9.02)
NF_t	0.061	0.057	0.021
	(1.47)	(1.42)	(0.61)
DUM_1990	0.007	-0.000	-0.012
	(0.25)	(-0.01)	(-0.42)
DUM_2000	-0.047	-0.056	-0.065
	(-0.87)	(-1.05)	(-1.34)
CONSTANT	0.073**	0.078***	0.086***
	(2.52)	(2.63)	(2.98)
No. of obs.	108,357	108,357	108,357
Adj. R^2	0.190	0.193	0.207

Table 4: What Caused the Increase in the Value of Cash?

Table 4 examines the economic determinants of the value of cash over time. IPO dummy (IPO_DUMMY) equals to 1 if a firm has gone public within the prior five years. R&D (RD) is measured by an indicator variable that equals to 1 if a firm-year observation has above-median R&D among all non-zero R&D firms. Cash-flow volatility (CASH-FLOW_VOLATILITY) is the cash-flow standard deviation of a firm based on its 2-digit SIC code. Hence, for an industry in any given year, we compute the cash-flow standard deviations of all firms in the industry from the past 20 years, and take the industry average cash-flow standard deviation. We require all firms to have at least five observations. Financial constraints are measured by the Hadlock and Pierce (2010) Size-and-Age index (SA_INDEX). We sort firms into SA quintiles for each decade based on its lagged value of SA index. Credit spread (CREDIT_SPREAD) is the annual yield spread between Aaa bonds and Baa bonds. HHI is the product market concentration measure based on Fama–French 48-industry classifications. Multi-segment (MULTISEG_DUMMY) is an indicator that equals to 1 if a firm has three or more segments. Investor sentiment (SENTIMENT) is measured following Baker et al. (2012). In Panel A, we examine the impact of these determinants one at a time. In Panel B, we include all determinants in a single regression. All variables from model 3 of Table 3 are included as control variables for all specifications, but their coefficient estimates are suppressed for brevity. All variables have been winsorized at the top and bottom one percentile to mitigate the impact of extreme outliers. The sample period is from 1980 to 2009. In parentheses are *t*-statistics s based on standard errors that are robust to heteroskedasticity and firm and year clustering. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table 4 (continued)

Panel A. The Impact of Individual Determinants

	Determinant							
Variables	IPO 1	R&D 2	Cash-Flow Volatility 3	Financial Constraints 4	Credit spread 5	HHI 6	Multi-segment 7	Sentiment 8
ΔC_t	1.013***	0.871***	1.066***	0.797***	0.994***	1.068***	1.022***	1.008***
	(10.58)	(9.05)	(7.53)	(4.39)	(5.60)	(11.70)	(10.61)	(10.73)
ΔC_t × DUM_1990	0.821***	0.572***	0.153	0.642*	0.556	0.622***	0.793***	0.646***
	(2.64)	(2.72)	(0.62)	(1.74)	(0.99)	(2.64)	(2.70)	(3.04)
ΔC_t × DUM_2000	1.023***	1.101***	0.776**	0.743*	0.553*	0.960**	1.058**	1.014***
	(2.58)	(2.82)	(2.03)	(1.75)	(1.66)	(2.38)	(2.53)	(2.68)
ΔC_t × IPO_DUMMY	0.339***	0.276***	0.354***	0.260***	0.338***	0.330***	0.327***	0.342***
	(4.93)	(4.35)	(5.23)	(4.19)	(5.39)	(4.82)	(4.75)	(4.89)
ΔC_t × IPO_DUMMY × DUM_1990	0.329*	0.227	0.239	0.284	0.345*	0.350*	0.349*	0.317*
	(1.80)	(1.56)	(1.54)	(1.49)	(1.95)	(1.90)	(1.86)	(1.90)
ΔC_t × IPO_DUMMY × DUM_2000	0.270	0.308	0.268	0.181	0.271	0.279	0.257	0.254
	(1.31)	(1.58)	(1.29)	(0.98)	(1.25)	(1.36)	(1.26)	(1.23)
IPO_DUMMY	-0.063***	-0.058***	-0.061***	-0.040***	-0.063***	-0.062***	-0.059***	-0.063***
	(-4.42)	(-4.23)	(-4.20)	(-3.23)	(-4.45)	(-4.41)	(-4.16)	(-4.35)
ΔC_t × Determinant		0.612***	-1.337	0.070*	0.015	0.487	-0.127***	0.050*
		(7.26)	(-0.60)	(1.73)	(0.14)	(1.54)	(-2.67)	(1.74)
ΔC_t × Determinant × DUM_1990		0.800**	6.988**	0.259***	0.372	-0.914*	0.305**	0.103*
		(2.04)	(2.38)	(2.80)	(0.83)	(-1.71)	(2.13)	(1.68)
ΔC_t × Determinant × DUM_2000		-0.437***	2.291	0.110*	0.472**	-0.433	-0.047	-0.109
		(-3.19)	(1.03)	(1.73)	(1.98)	(-0.39)	(-0.31)	(-1.07)
Determinant		-0.067***	-0.436***	-0.019***	0.044	-0.190***	0.034***	0.004
		(-2.65)	(-4.82)	(-2.77)	(0.58)	(-3.14)	(4.26)	(0.49)
	All control variables from column 3 of Table 3 and an intercept are included.							
No. of obs.	108,357	108,357	108,357	108,339	108,339	108,336	108,357	108,357
Adj. R^2	0.210	0.215	0.213	0.213	0.212	0.210	0.211	0.212

Table 4 (continued)

Panel B. The Impact of All Determinants Simultaneously

	Determinant							
	IPO	R&D	Cash-Flow Volatility	Financial Constraints	Credit spread	HHI	Multi-segment	Sentiment
Variables	1	2	3	4	5	6	7	8
ΔC_t	0.760***							
	(5.47)							
ΔC_t × DUM_1990	-0.425							
	(-1.07)							
ΔC_t × DUM_2000	0.451							
	(1.07)							
ΔC_t × Determinant	0.207***	0.496***	-1.311	0.084**	-0.022	0.472	-0.066	0.045*
	(3.00)	(5.37)	(-1.06)	(2.18)	(-0.27)	(0.80)	(-1.14)	(1.83)
ΔC_t × Determinant × DUM_1990	0.250*	0.682*	3.672***	0.137*	0.647*	-0.880*	0.260**	0.098**
	(1.69)	(1.76)	(2.76)	(1.84)	(1.75)	(-1.88)	(2.48)	(2.04)
ΔC_t × Determinant × DUM_2000	0.226	-0.419***	1.055	0.126**	0.569***	-7.407	-0.054	-0.145
	(1.07)	(-3.21)	(1.29)	(2.24)	(2.68)	(-1.10)	(-0.38)	(-1.34)
Determinant	-0.040***	-0.042*	-0.339***	-0.013*	0.041	-0.104*	0.018**	0.002
	(-3.39)	(-1.67)	(-3.12)	(-1.83)	(0.54)	(-1.66)	(2.13)	(0.20)
	All control variables from column 3 of Table 3 and an intercept are included.							
No. of obs.								108,318
Adj. R^2								0.221

Table 5: The Speed of Adjustment (SOA) in Cash Holdings

Table 5 shows the OLS regression results of the speed of adjustment (SOA) in cash holdings. In the following regressions, SOA is $[1 - (\gamma_1 + \gamma_2 \times \text{EX\_CASH}_{i,t-1})]$. In Panel A, models 1 and 2 are cross-sectional regressions, and models 3 and 4 interact with decade indicators. The estimation equation is:

$\text{CASH}_{it} = \gamma_0 + \gamma_1 \times \text{CASH}_{i,t-1} + \gamma_2 \times \text{EX\_CASH}_{i,t-1} \times \text{CASH}_{i,t-1} + \gamma_3 \times \text{EX\_CASH}_{i,t-1} + \beta * x_{i,t} + \epsilon_{i,t}$,

where $x_{i,t}$ consists of a set of the following control variables. CASH-FLOW_VOLATILITY is defined as standard deviations of cash-flow-to-total-assets ratio of 2-digit SIC industries using past 20 years observations. MB is defined as the book value of assets minus the book value of equity plus the market value of equity over the book value of assets. FIRM_SIZE is defined as the logarithm of the book value of total assets in 2004 dollars. CASH_FLOW is defined as earnings after interest, dividends, and taxes but before depreciation over the book value of assets. NET_WORKING_CAPITAL is defined as working capital minus cash over the book value of assets. CAPITAL_EXPENDITURE is defined as capital expenditures over the book value of assets. L is defined as long-term debt plus debt in current liabilities over the book value of assets. R&D (RD) is defined as research and development expenses over sales. DIVIDEND_DUMMY equals 1 if a firm pays a common dividend in a given year. ACQUISITIONS are defined as acquisition expenditures over the book value of assets. In models 1 and 3, EX_CASH_{it-1} is the excess-cash ratio, which is the difference between the actual cash ratio and the predicted cash ratio. In models 2 and 4, EX_CASH_{it-1} is excess-cash quartile. The bottom quartile takes the value 1, and the top quartile takes the value 4. To form the quartiles, we first sort firms into two sub-groups based on whether a firm has positive or negative excess cash. For each of the two sub-groups, we again sort firms into two halves. Hence, the top quartile group includes firms with the highest excess cash. Panel B reports the SOA for each quartile in each decade as well as the change in SOA over time (the bottom row) and the difference in SOA across quartiles (the rightmost column). All variables have been winsorized at the top and bottom one percentile to mitigate the impact of extreme outliers. The final sample covers all Compustat firm-year observations with no missing values on requisite variables. The sample period is from 1980 to 2009. Financial firms (SIC codes between 6000 and 6999) and utility firms (SIC codes between 4900 and 4999) are excluded from the sample. In parentheses are *t*-statistics based on standard errors that are robust to heteroskedasticity and firm and year clustering. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table 5 (continued)
Panel A. SOA Regressions

Variables	EX_CASH			
	Continuous Excess-Cash Ratio 1	Excess-Cash Quartile 2	Continuous Excess-Cash Ratio 3	Excess-Cash Quartile 4
LAGGED_CASH	0.627***	0.689***	0.515***	0.503***
	(30.47)	(37.64)	(20.42)	(10.41)
LAGGED_CASH × DUM_1990			0.095***	0.171***
			(3.95)	(3.17)
LAGGED_CASH × DUM_2000			0.147***	0.214***
			(6.74)	(4.25)
LAGGED_CASH × EX_CASH	-0.108***	-0.007**	0.121**	0.030***
	(-5.40)	(-2.10)	(2.38)	(2.80)
LAGGED_CASH × EX_CASH × DUM_1990			-0.252***	-0.037***
			(-3.73)	(-3.04)
LAGGED_CASH × EX_CASH × DUM_2000			-0.252***	-0.034***
			(-4.31)	(-2.93)
EX_CASH	0.173***	0.012***	0.174***	0.011***
	(10.26)	(18.96)	(9.97)	(17.10)
CASH-FLOW_VOLATILITY	0.257***	0.202***	0.295***	0.243***
	(7.92)	(5.97)	(12.51)	(10.02)
MB	0.008***	0.007***	0.009***	0.008***
	(16.10)	(13.56)	(17.73)	(15.51)
FIRM_SIZE	-0.003***	-0.002***	-0.003***	-0.002***
	(-7.63)	(-6.25)	(-8.36)	(-7.01)
CASH_FLOW	0.036***	0.041***	0.037***	0.042***
	(8.02)	(8.43)	(8.49)	(8.80)
NET_WORKING_CAPITAL	-0.136***	-0.126***	-0.138***	-0.128***
	(-29.89)	(-31.95)	(-30.28)	(-32.80)
CAPITAL_EXPENDITURE	-0.325***	-0.316***	-0.325***	-0.317***
	(-33.40)	(-31.34)	(-38.64)	(-35.75)
L	-0.140***	-0.126***	-0.143***	-0.127***
	(-17.79)	(-22.42)	(-18.16)	(-23.79)
RD	0.018***	0.014***	0.016***	0.014***
	(6.12)	(6.44)	(6.04)	(6.31)
DIVIDEND_DUMMY	-0.011***	-0.010***	-0.012***	-0.011***
	(-9.74)	(-9.20)	(-11.38)	(-10.22)
ACQUISITIONS	-0.356***	-0.360***	-0.352***	-0.356***
	(-15.86)	(-16.17)	(-15.47)	(-15.84)
DUM_1990			-0.019***	-0.018***
			(-8.47)	(-7.08)
DUM_2000			-0.022***	-0.021***
			(-9.51)	(-8.15)
CONSTANT	0.114***	0.076***	0.128***	0.089***
	(19.76)	(16.41)	(28.61)	(25.37)
No. of obs.	107,022	107,022	107,022	107,022
Adj. R^2	0.782	0.782	0.784	0.783

Table 5 (continued)

Panel B. The Speed of Adjustment (SOA) for Quartiles of Excess Cash Over Time

	Bottom Quartile 1	2	3	Top Quartile 4	Difference in SOA (Bottom minus Top Quartile)
1980	0.467	0.437	0.407	0.377	0.09
1990	0.333	0.340	0.347	0.354	-0.021
2000	0.287	0.291	0.295	0.299	-0.012
Change in SOA over time	-0.180	-0.146	-0.112	-0.078	

Table 6: Financial Constraints and the Speed of Adjustment (SOA) in Cash Holdings Over Time

Table 6 examines whether the speed of adjustment (SOA) in cash holdings is associated with financial constraints. We classify firms into less-constrained vs. more-constrained subgroups and run the specification of model 4 in Table 4 Panel A. The more-constrained subgroups are in even-numbered columns. Financial constraints are defined as above-median SA index, above-median WW index, firms with no bond credit rating or commercial paper credit rating, and firms that not pay dividends. The less-constrained subgroups are in odd-numbered columns. All variables have been winsorized at the top and bottom one percentile. The final sample covers all Compustat firm-year observations with no missing values on requisite variables. The sample period is from 1980 to 2009. Financial firms (SIC codes between 6000 and 6999) and utility firms (SIC codes between 4900 and 4999) are excluded from the sample. In parentheses are *t*-statistics based on standard errors that are robust to heteroskedasticity and firm and year clustering. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

	Financial Constraints									
	SA Index		WW Index		Bond		Commercial Paper		Dividend	
	Below-Median	Above-Median	Below-Median	Above-Median	Rated	Unrated	Rated	Unrated	Payer	Non-payer
Variables	1	2	3	4	5	6	7	8	9	10
LAGGED_CASH	0.599***	0.529***	0.629***	0.538***	0.710***	0.547***	0.658***	0.584***	0.544***	0.566***
	(13.22)	(11.97)	(13.20)	(13.47)	(14.15)	(16.17)	(14.50)	(12.71)	(12.59)	(14.70)
LAGGED_CASH × DUM_1990	-0.029	0.146***	0.009	0.126***	-0.063	0.132***	-0.070	0.101**	-0.105***	0.115**
	(-0.60)	(2.97)	(0.17)	(2.89)	(-1.16)	(3.38)	(-1.29)	(2.04)	(-2.67)	(2.56)
LAGGED_CASH × DUM_2000	0.101*	0.176***	0.093*	0.163***	0.063	0.163***	-0.021	0.140***	-0.008	0.149***
	(1.89)	(3.93)	(1.66)	(4.11)	(0.98)	(4.98)	(-0.41)	(3.03)	(-0.14)	(3.71)
LAGGED_CASH × EX_CASH	0.012	0.020**	0.004	0.019**	-0.017	0.027***	-0.003	0.018	0.032***	0.009
	(1.02)	(2.15)	(0.33)	(2.37)	(-1.29)	(2.69)	(-0.21)	(1.38)	(3.64)	(1.06)
LAGGED_CASH × EX_CASH × DUM_1990	0.013	-0.031***	0.005	-0.027***	0.022	-0.036***	0.021	-0.027**	0.030***	-0.020**
	(1.07)	(-2.79)	(0.41)	(-2.87)	(1.46)	(-3.29)	(1.60)	(-1.96)	(3.19)	(-2.02)
LAGGED_CASH × EX_CASH × DUM_2000	-0.008	-0.024**	-0.007	-0.021**	0.001	-0.030***	0.016	-0.022*	0.012	-0.014
	(-0.65)	(-2.35)	(-0.55)	(-2.37)	(0.03)	(-2.95)	(1.12)	(-1.70)	(0.94)	(-1.45)
EX_CASH	0.010***	0.009***	0.010***	0.010***	0.008***	0.010***	0.010***	0.010***	0.010***	0.010***
	(11.79)	(10.84)	(11.82)	(11.92)	(7.96)	(11.05)	(7.64)	(14.22)	(8.77)	(14.37)
	All control variables from column 4 of Table 4 Panel A and an intercept are included.									
No. of obs.	53,510	53,512	53,137	53,139	32,330	51,124	12,566	70,888	39,962	67,060
Adj. R^2	0.781	0.768	0.793	0.771	0.747	0.801	0.741	0.798	0.786	0.773

Table 7: The Effect of Block Holdings on the Value of Cash Over Time

Table 7 examines whether institutional block holdings can explain the increase in the value of cash. Based on Panel B in Table 4, we add institutional block holdings (5%_BLOCK) as a measure that is the sum of all institutional holdings that are greater than 5% of a firm's equity. The data are from Thomson Reuters Institutional Holdings (13F) Database. The subscripts *t* indicates that the value of the variable is at the end of fiscal year *t*. All variables have been winsorized at the top and bottom one percentile to mitigate the impact of extreme outliers. The sample period is from 1980 to 2009. In parentheses are *t*-statistics based on standard errors that are robust to heteroskedasticity and firm and year clustering. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Variables	Inst. Block
ΔC_t	0.777***
	(2.59)
ΔC_t × DUM_1990	-0.229
	(-0.49)
ΔC_t × DUM_2000	0.496
	(1.20)
ΔC_t × 5%_BLOCK$_t$	-0.049
	(-0.11)
ΔC_t × 5%_BLOCK$_t$ × DUM_1990	0.957*
	(1.75)
ΔC_t × 5%_BLOCK$_t$ × DUM_2000	-0.883
	(-1.53)
5%_BLOCK$_t$	-0.149*
	(-1.85)
All other variables from Table 4 Panel B and an intercept are included.	
No. of obs.	103,738
Adj. R^2	0.223